Assessing Upslope Range Shifts and Methodological Bias in Vegetation Resurveys of E.C. Manning Provincial Park

THE UNIVERSITY OF BRITISH COLUMBIA

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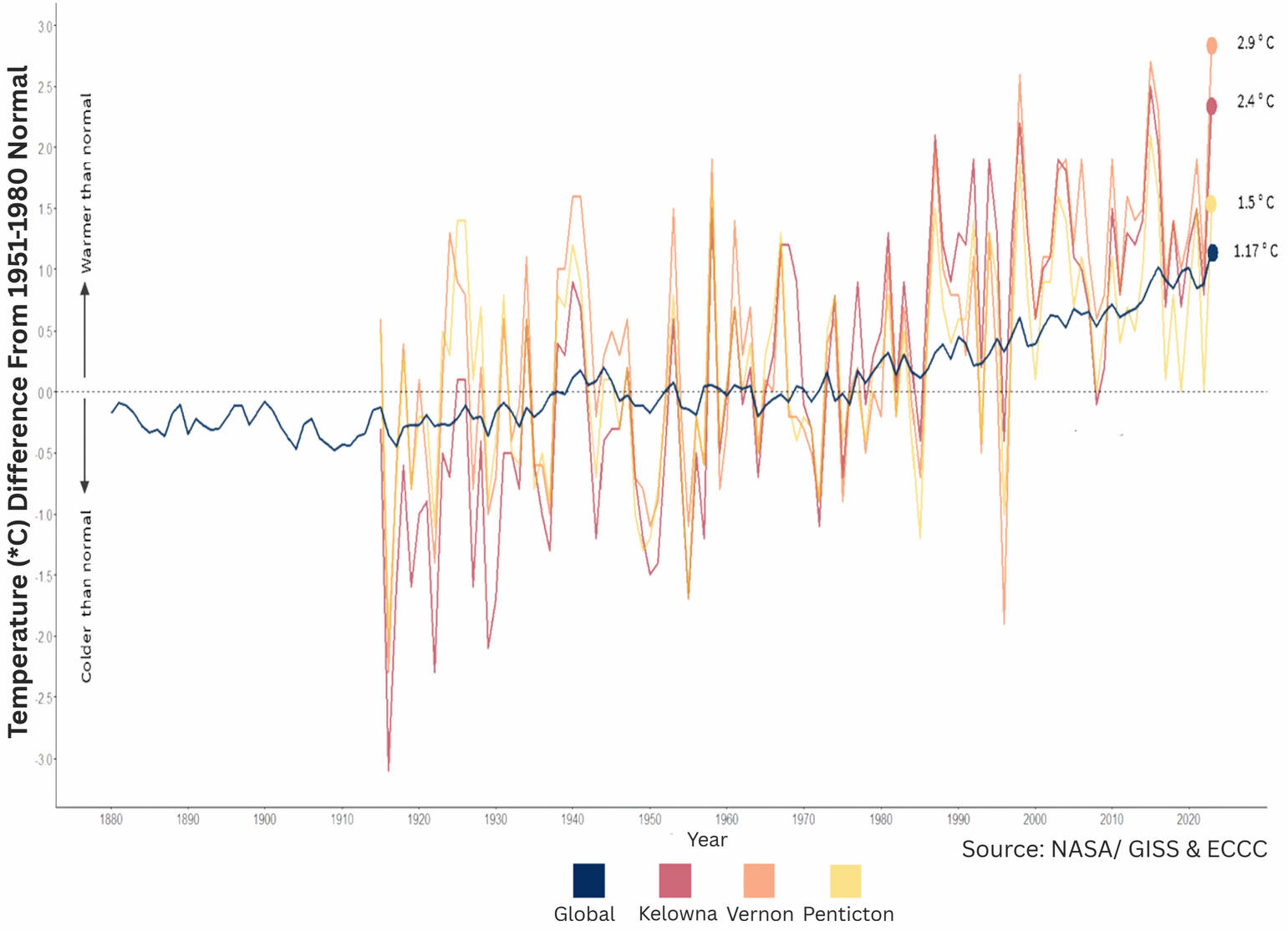
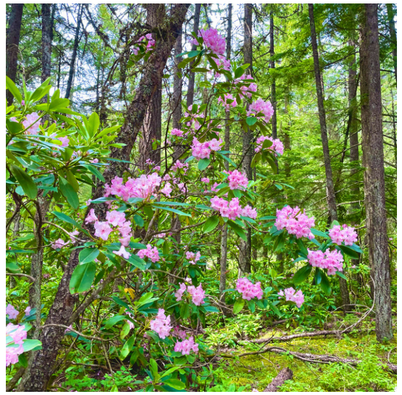
Mean annual temperatures in the Coast-Interior Transitional region and Okanagan are warming faster than the global average

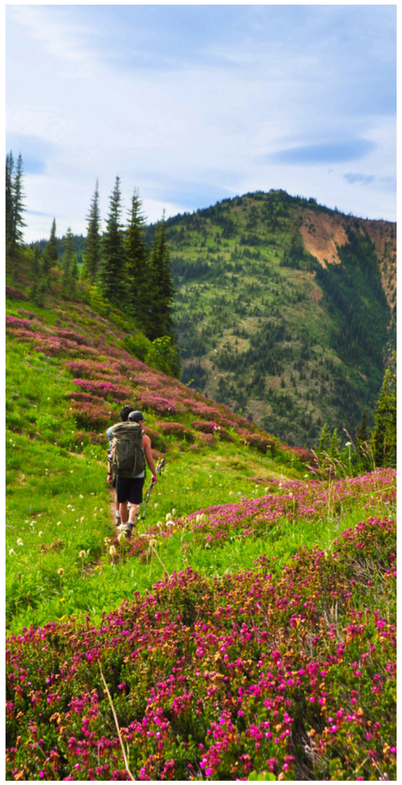


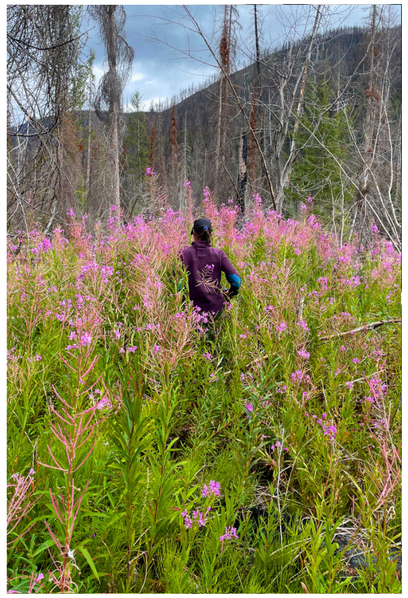
Source: NASA/ GISS & ECCC

The plant ecologist's question of the century:

How are plant species and communities responding to climate change?







Plant species shift their ranges in response to climate change

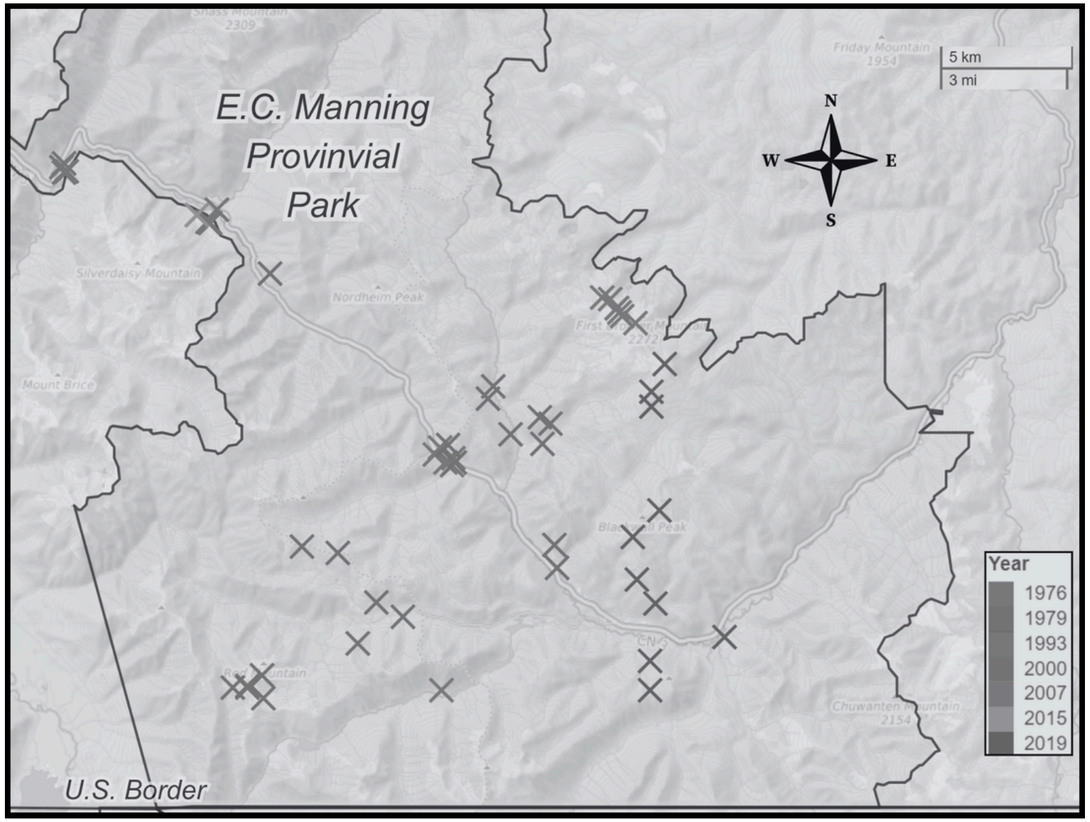
Range Shifts

- The movement of a species distribution across latitudinal or elevational gradients

- Examples: Range contractions, range expansions, shifts in the peak probability of occurrence, stasis

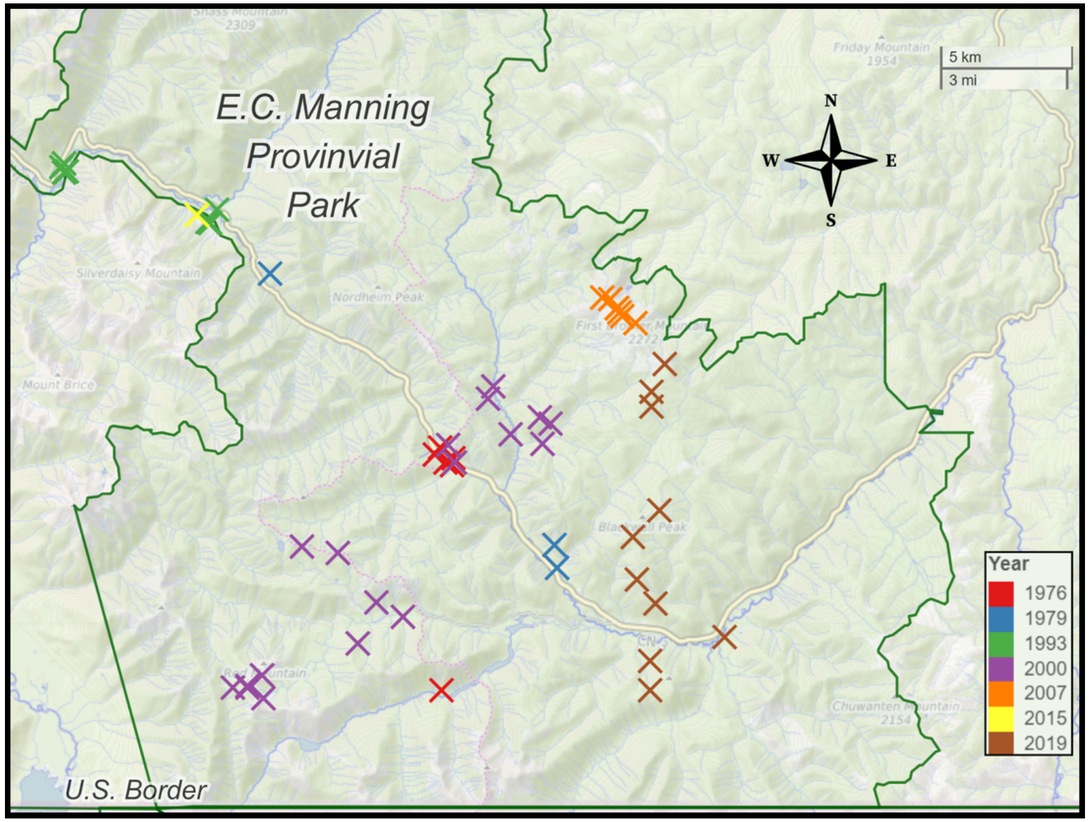
We can track changes to plant communities with resurvey projects

Historical Surveys



- Historically conducted to make maps or identify prime areas for agriculture, forestry, or mining
- Data usually contain plant species' presences and abundances

Resurvey from 2025



- Use historical data as a baseline to measure how plants are responding to recent climate change (Vellend et al. 2013)

Source: OpenStreetMap, ClimateNA

Challenges associated with resurvey studies and gaps in the literature

Problem: Historical sites can be difficult to relocate (relocation error)

- Historical GPS is unreliable
- **Pseudo-turnover**: artificial changes in species turnover observed in resurvey studies caused by relocation error

Gap 1: Few studies have quantified their own relocation error and pseudo-turnover in resurvey projects

Gap 2: There has yet to be a resurvey study conducted in the Coast-Interior Transitional Zone and the Okanagan Region

Key study question and objectives

Key question:

How have plant communities in E.C. Manning Park changed over the past fifty years along an elevational gradient, and how does relocation uncertainty affect our ability to detect those changes?

Objectives:

1. **Evaluate the influence of relocation error on estimates of change**

2. **Assess directional community responses to climate change in Manning Park**

Study System

E.C. Manning Provincial Park

- Protected provincial park **since 1941**, a game reserve as far back as 1931
- Spans an elevational range of **~600-2300 m**
- 117 historical vegetation sites available from **1976, 1979, 1993, 2000, 2007, 2015, 2019**
- Resurveyed **52 sites** of the sites available

Biogeoclimatic Ecosystem Classification (BEC) database

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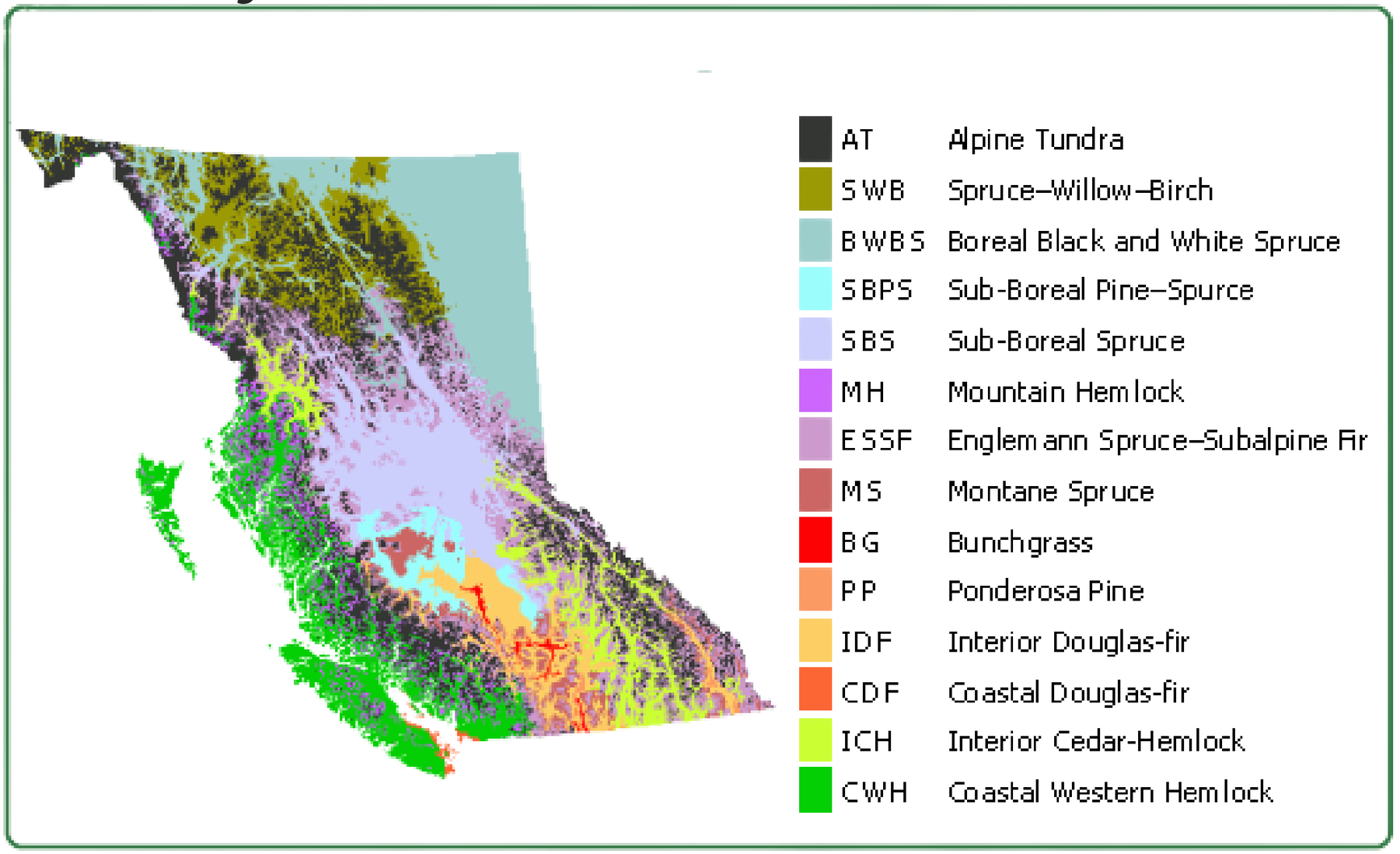


- Used to make maps that classify ecosystems into BGC zones, subzones, and even more subzones

Biogeoclimatic Ecosystem Classification (BEC) database

Data provided by BEC:

Species presences and estimated abundances (% cover)

Parameters to help site relocation:

Latitude, longitude, elevation

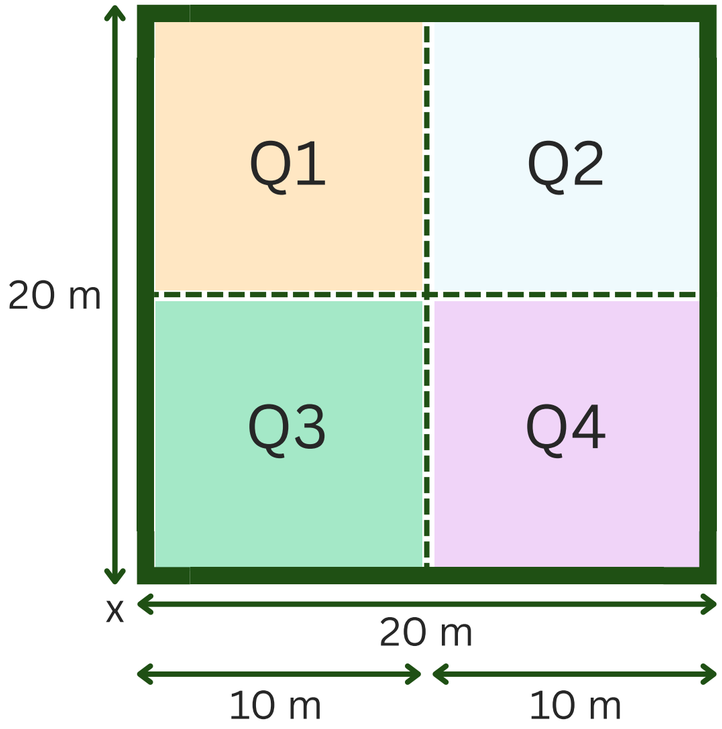
Slope aspect

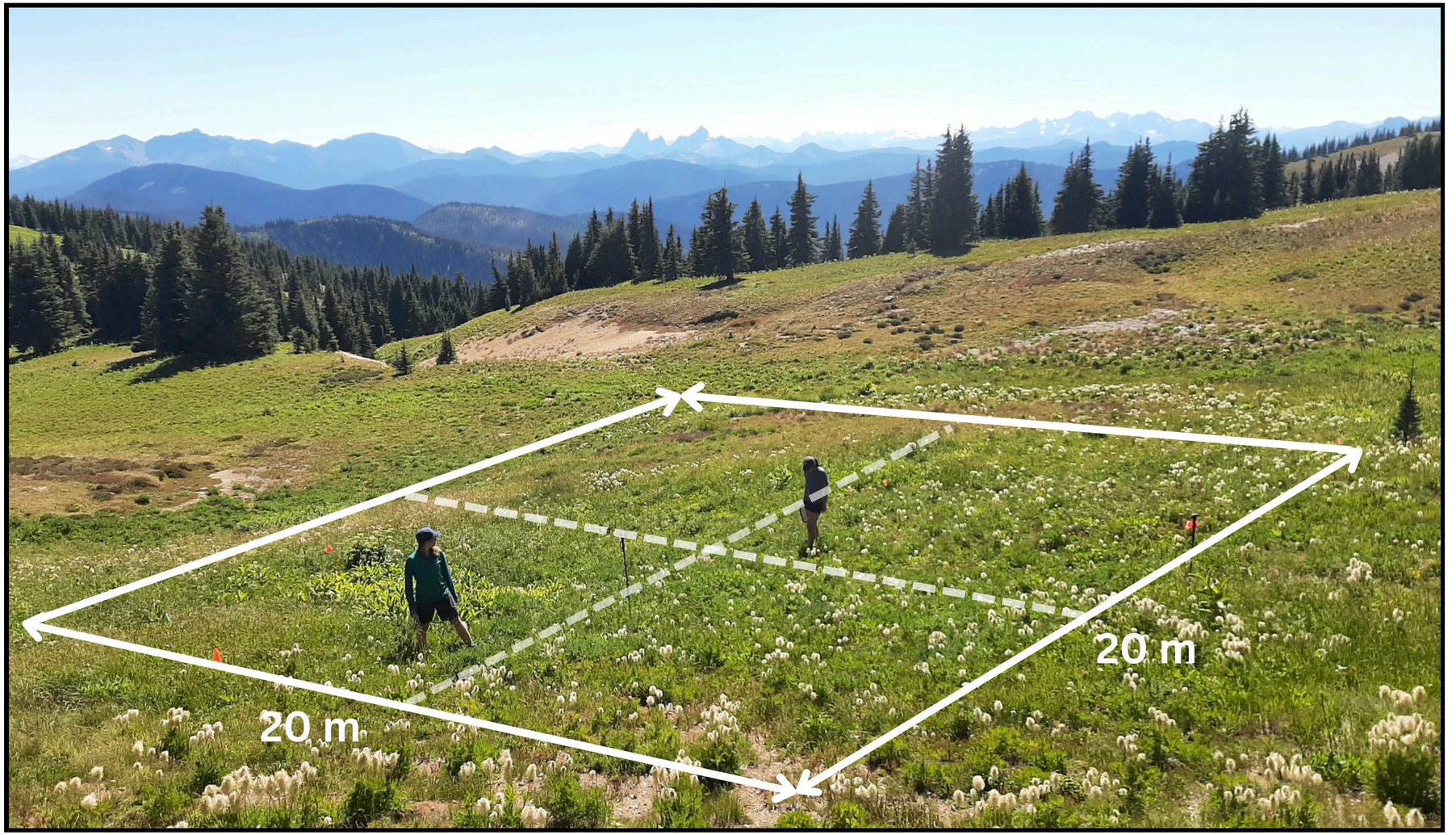
Slope gradient %

Mesoslope position

Site notes

Field sampling protocol for plots from 1976-2000 (n = 34)





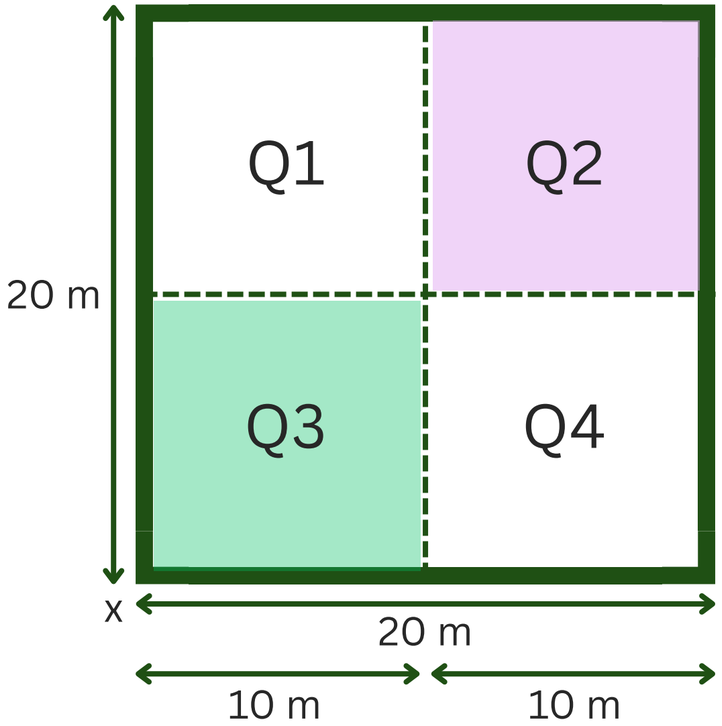
Methods & Results

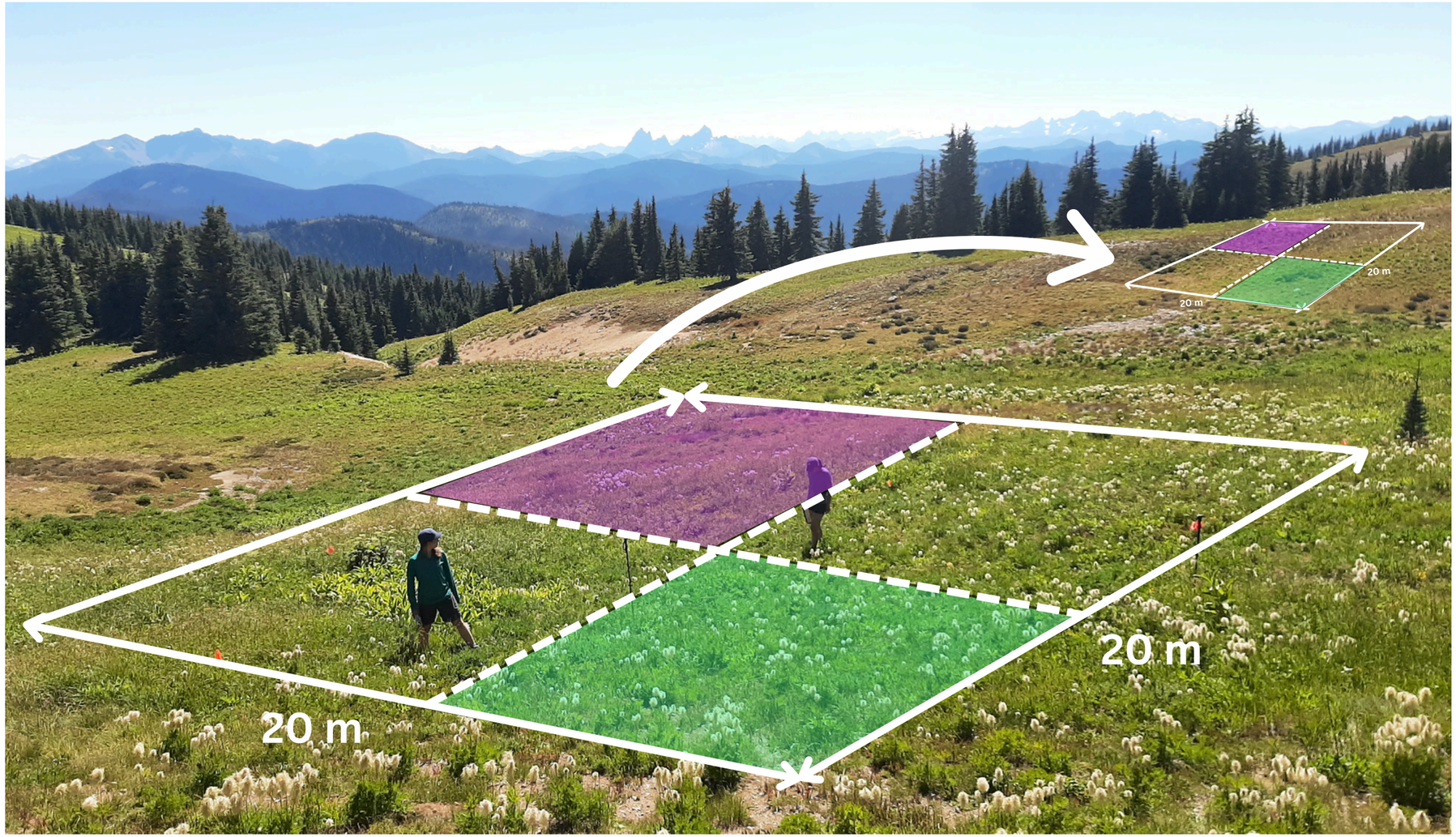
1. Evaluate the relocation error

2. Assess elevational response of plants

Relocation error analysis methods & results

Field sampling protocol for 2019 plots (n = 10)

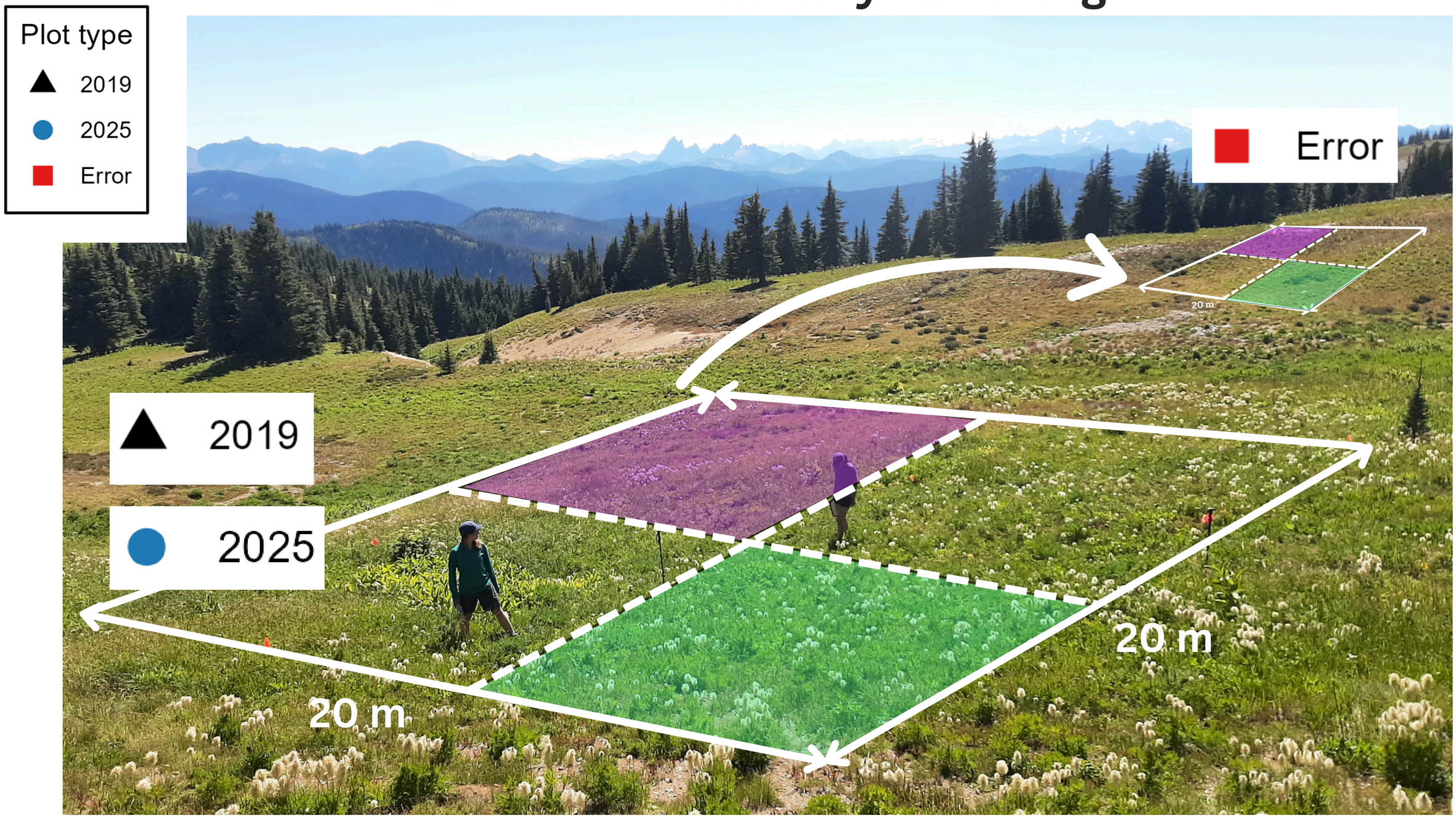




Criteria for relocating a plot:

- Elevation, % slope, aspect, mesoslope, community composition
- Represents a worst-case, but plausible relocation error scenario

Located ~60 m away on average



Relocation error analyses

1. **Compared SR, Shannon's diversity, Simpson's diversity**
    a. Friedman test

2. **Compare Bray-Curtis & Jaccard dissimilarity**
    a. Pairwise PERMANOVA of dissimilarity
        i. 2019 vs. $2025_{Correct}$
        ii. 2019 vs. 2025_{Error}
        iii. $2025_{Correct}$ vs. 2025_{Error}
    b. Non-metric multidimensional scaling to visualize (NMDS)

Hypotheses & predictions

1. **Hypothesis**
    - Relocation error inflates estimates of vegetation change (pseudo-turnover)

2. **Predictions:**
    - Pseudo-turnover: 2025_{Error} dissimilar from $2025_{Correct}$
    - Metric sensitivity: Bray-Curtis shows greater change than Jaccard

Results

Species Richness, Shannon, and Simpson did not significantly differ among plots

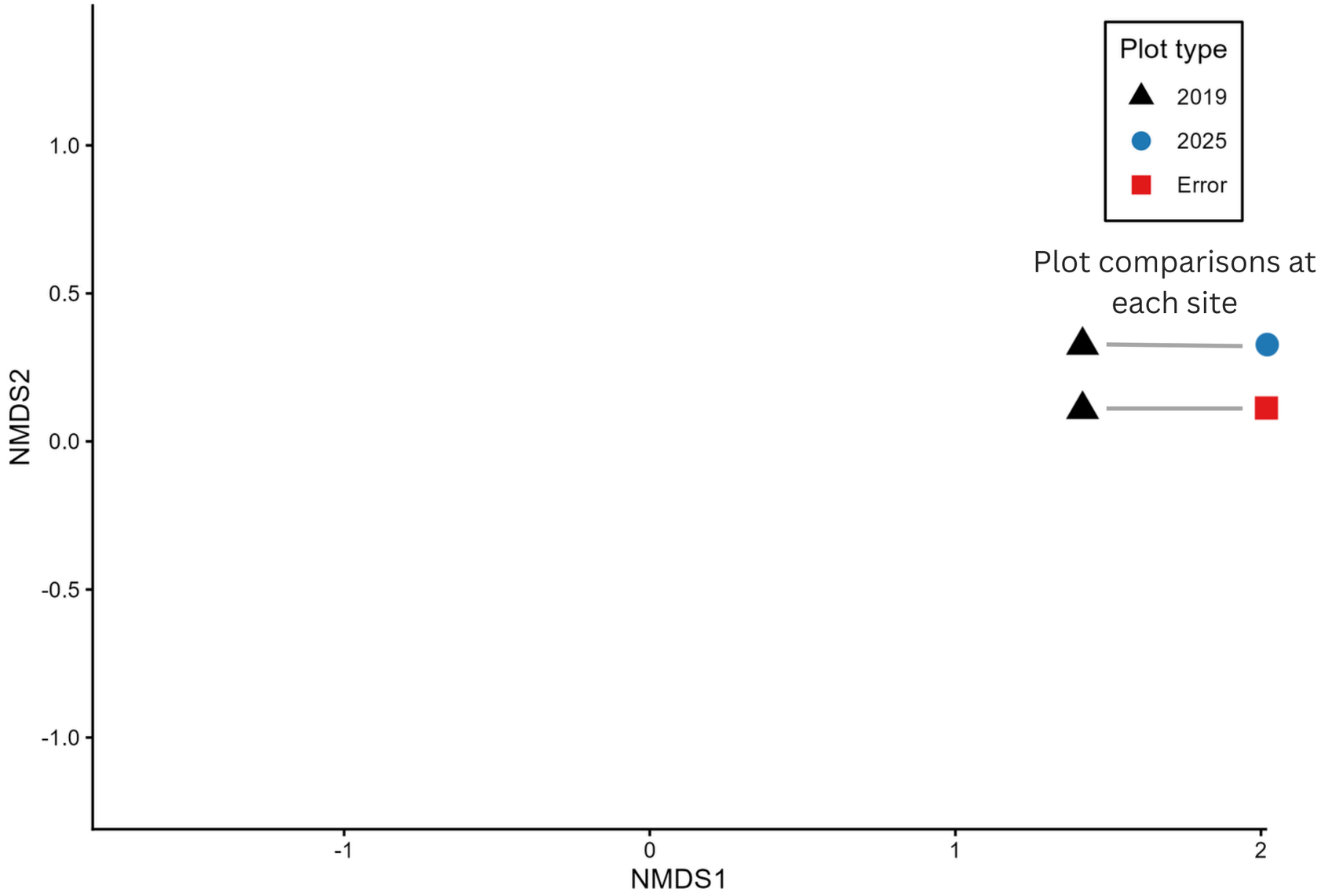
Table 6. Differences in species richness, Shannon's diversity, and Simpson's diversity among treatments (2019, $2025_{Correct}$, and 2025_{Error} plots).

Metric	Comparison / Test	Statistic	p-value
Species richness	2019 vs $2025_{Correct}$	V = 14.5	**0.3738**
	2019 vs 2025_{Error}	V = 12	**0.2349**
	$2025_{Correct}$ vs 2025_{Error}	V = 14.5	**0.6725**
Shannon diversity	Friedman test (3 treatments)	χ^2 = 0.667	**0.7165**
Simpson diversity	Friedman test (3 treatments)	χ^2 = 2.667	**0.2636**

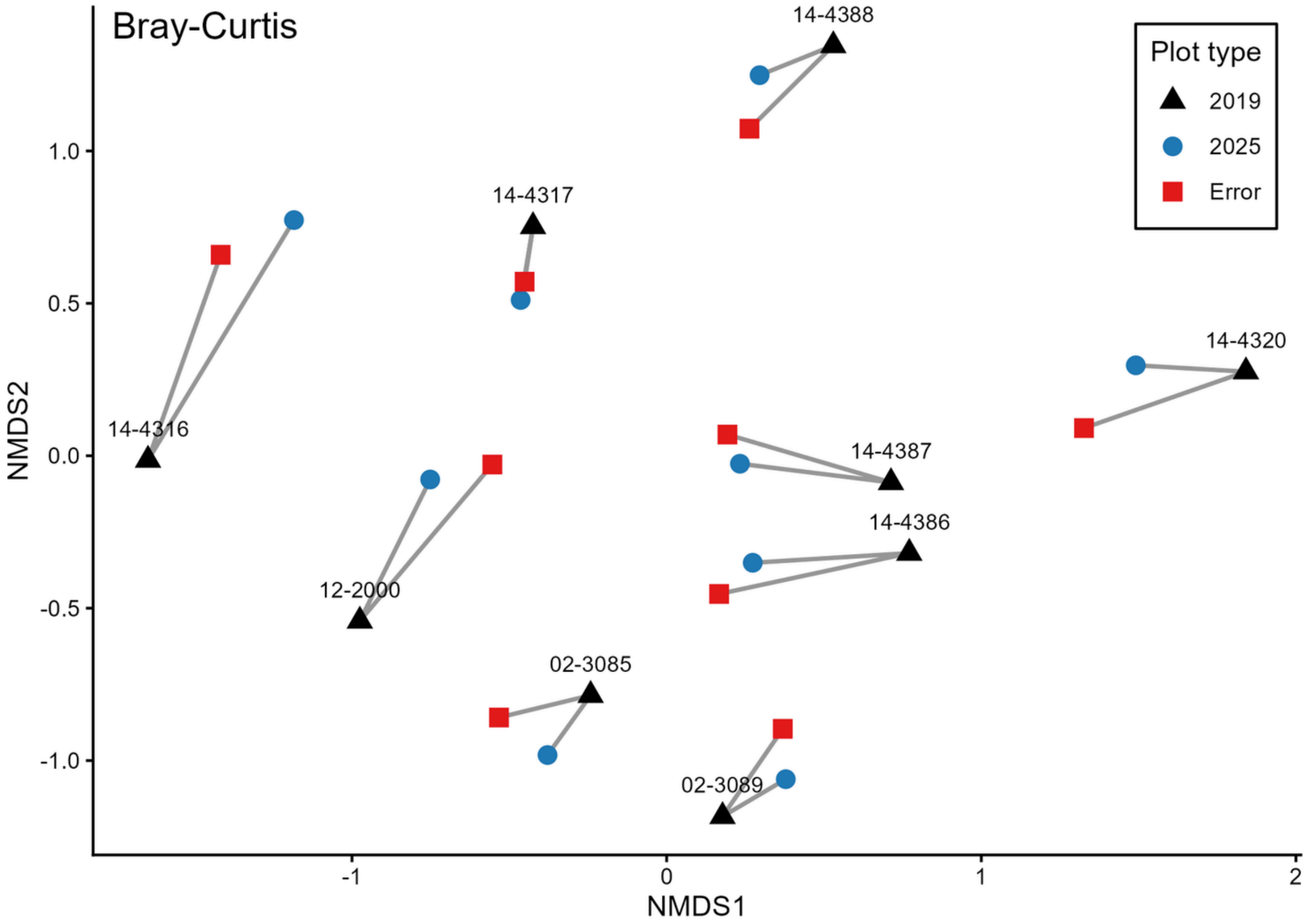
No significant differences:

- Survey effort is comparable
- Diversity metrics are robust to relocation error

Visuallizing dissimilarity across 2019 and 2025 plots

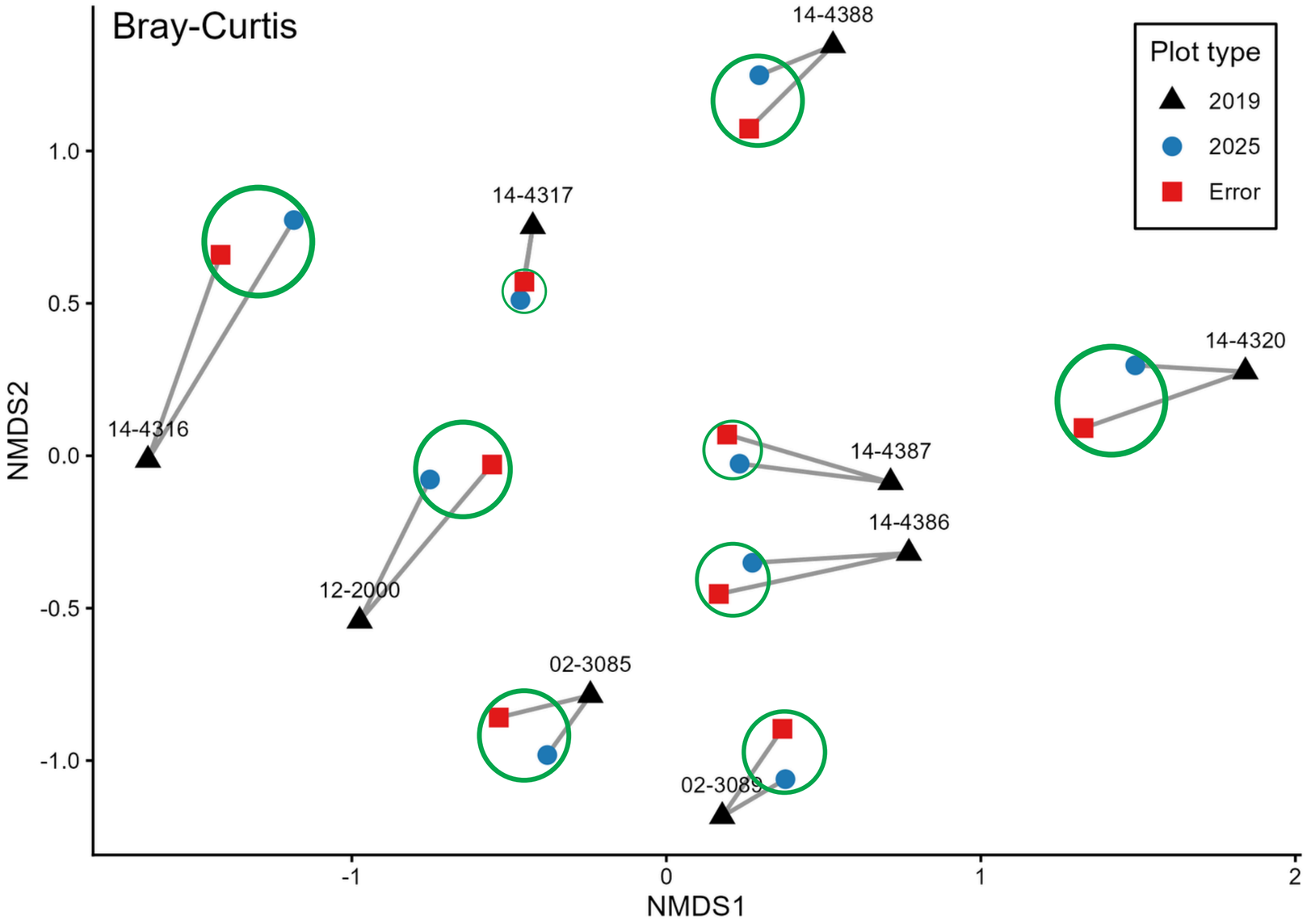


No significant difference between $2025_{Correct}$ and 2025_{Error}



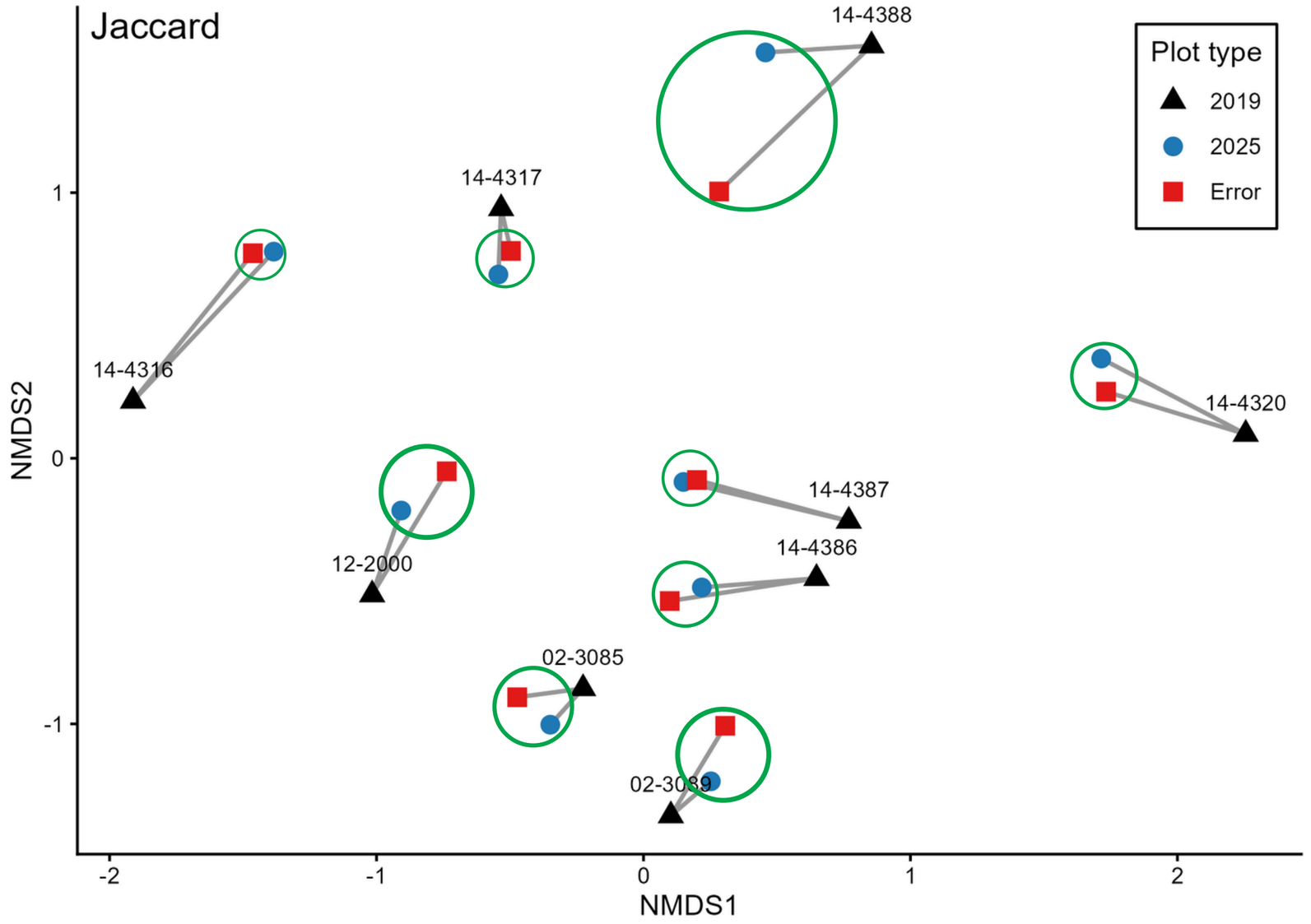
- $2025_{Correct}$ vs. 2025_{Error} (**p = 0.731**)
- Central convergence
- Same pattern across both metrics

No significant difference between $2025_{Correct}$ and 2025_{Error}



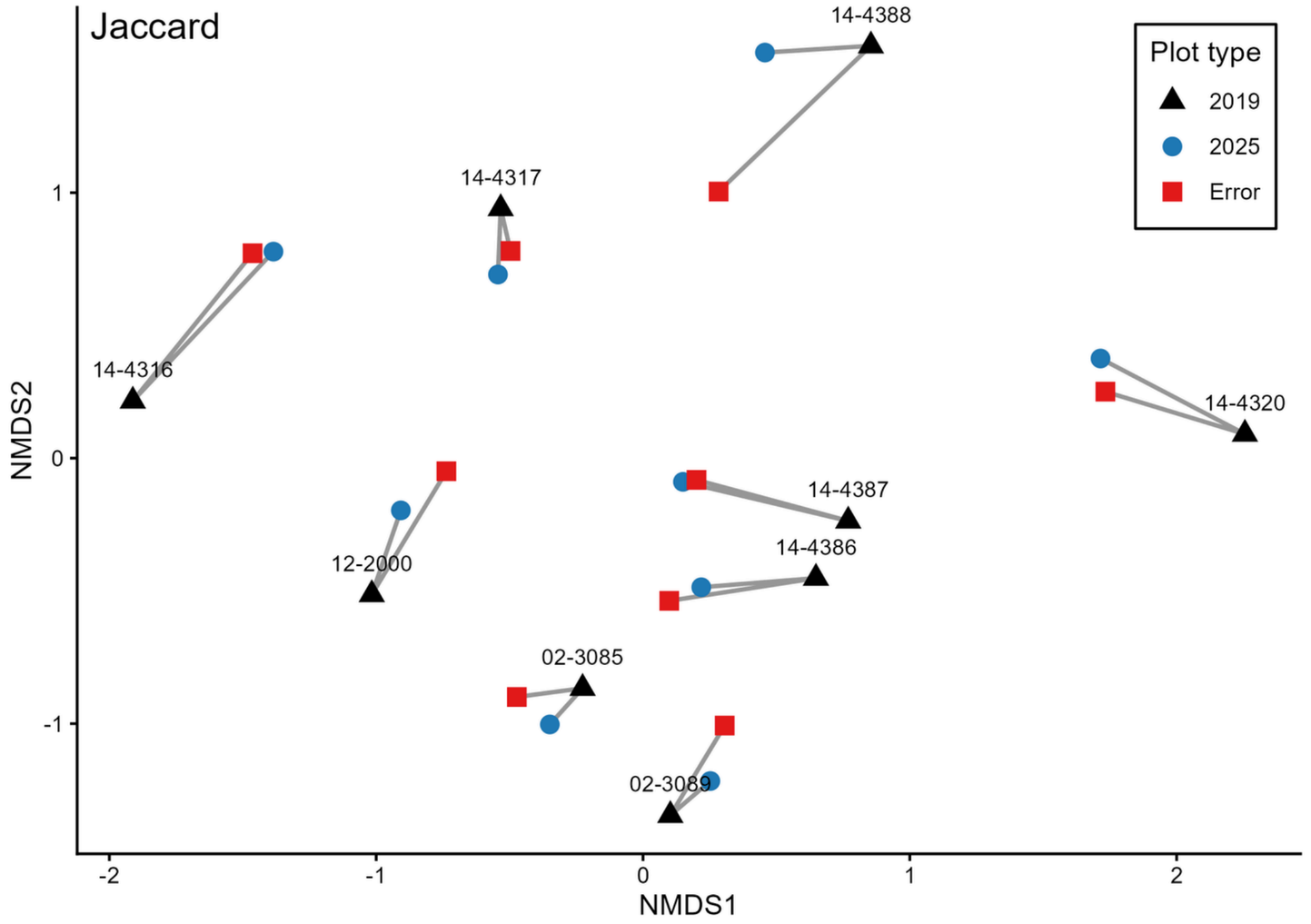
- $2025_{Correct}$ vs. 2025_{Error} (**p = 0.731**)
- Central convergence
- Same pattern across both metrics

No significant difference between $2025_{Correct}$ and 2025_{Error}



- $2025_{Correct}$ vs. 2025_{Error} (**p = 0.840**)
- Central convergence
- Same pattern across both metrics

No significant difference between $2025_{Correct}$ and 2025_{Error}



No significant difference:

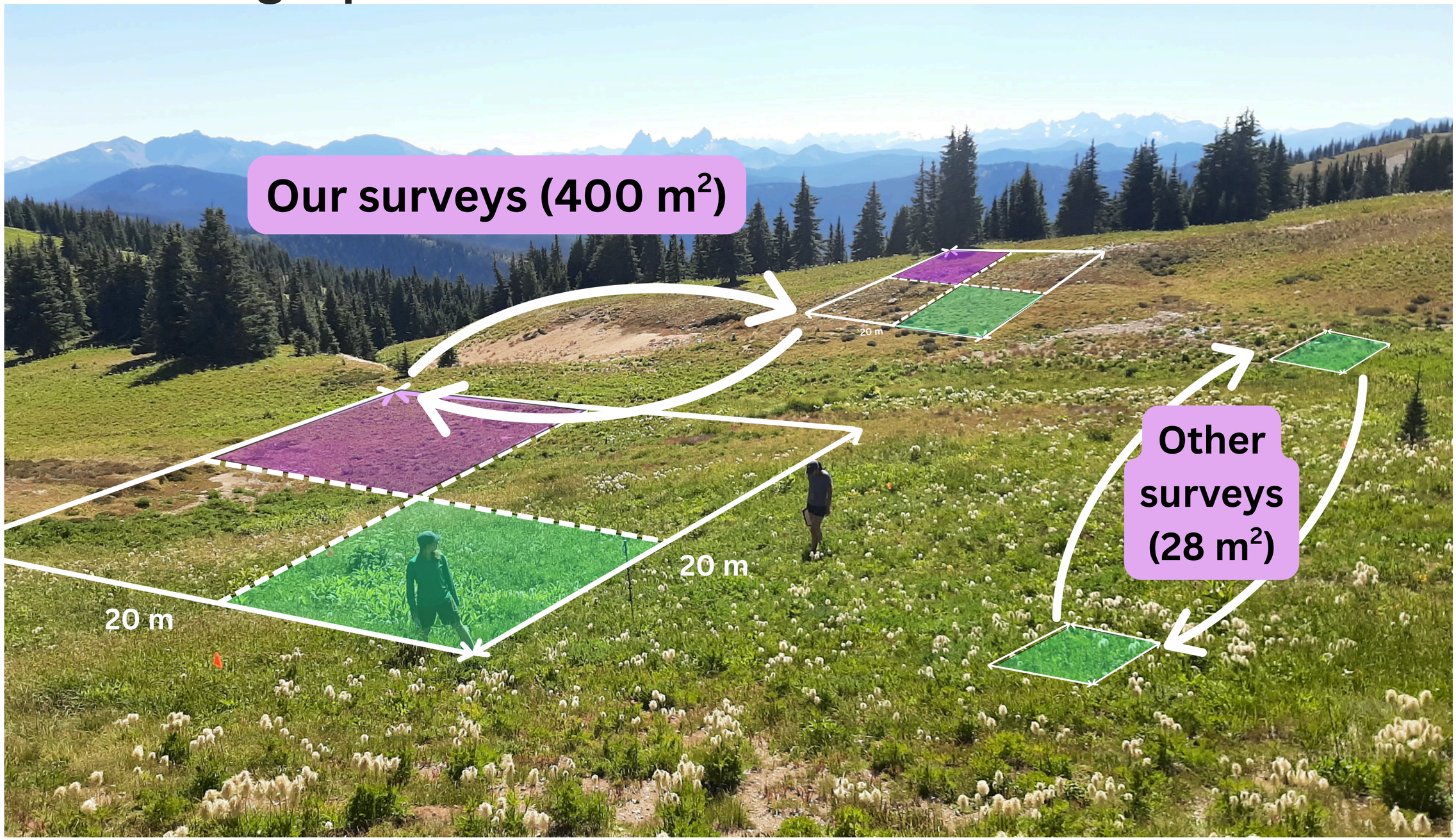
- No significant pseudo-turnover
- Estimates of change <u>are robust</u> to relocation error

In the literature:

- Aligns with Kopecký and Macek (2015) who found compositional metrics and species richness <u>were robust</u> to moderate relocation error

- In contrast, Boch et al. (2019) and Verheyen et al. (2018) found relocation error <u>did</u> introduce pseudo-turnover
  - Smaller plot size (28 m^2)

Larger plots are more robust to relocation error



Range shift analysis methods & results

Main Hypothesis Test:

Present ~ Time × (Elevation + Elevation^2)

Binomial GLMM

Model package in R: lme4

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Fixed effects:

- **Time: historical (1976-2000) vs. present (2025)**
- Elevation (m)
- Latitude
- Longitude

Main Hypothesis Test:

Present ~ Time × (Elevation + Elevation^2)

Binomial GLMM

Model package in R: lme4

Fixed effects:

- Time: historical (1979-2000) vs. present (2025)
- Elevation (m)
- Latitude
- Longitude

Potential random effects:

- Species
- Project ID
- Site Surveyor
- Site Number

Main Hypothesis Test:

Present ~ Time × (Elevation + Elevation^2)

Binomial GLMM

Model package in R: lme4

Fixed effects:

- Time: historical (1979-2000) vs. present (2025)
- Elevation (m)
- Latitude
- Longitude

Potential random effects:

- Species
- Project ID
- Site Surveyor
- Plot Number

Model selection using AIC:

- 26 candidate models

Final Model:

Present ~ Time × (Elevation + Elevation^2) + Latitude + (Elevation + Elevation^2 | Species) + (1 | Site Number)

Observations (n)	Sites (n)	Species (n)
2040	34	30

Post-hoc analysis:

$$\text{Present} \sim \text{Time} \times (\text{Elevation} + \text{Elevation}^2) \times \text{Functional group} + \text{Latitude} + (\text{Elevation} + \text{Elevation}^2 \mid \text{Species}) + (1 \mid \text{Site Number})$$

Herbs (n)	Shrubs (n)	Trees (n)
748	884	408

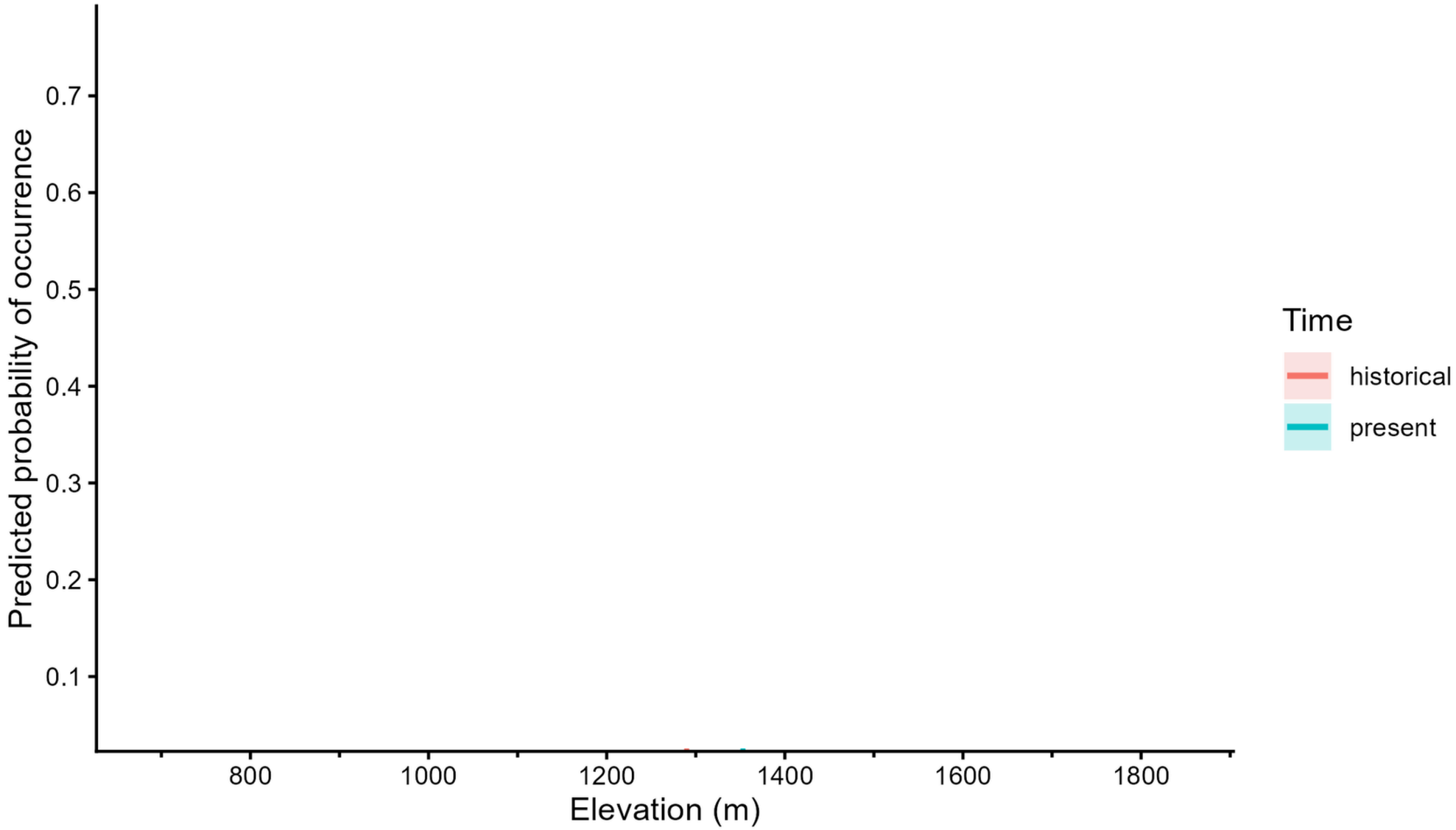
Hypotheses & predictions

1. **Hypothesis**
    - Warming drives upslope shifts in species distributions

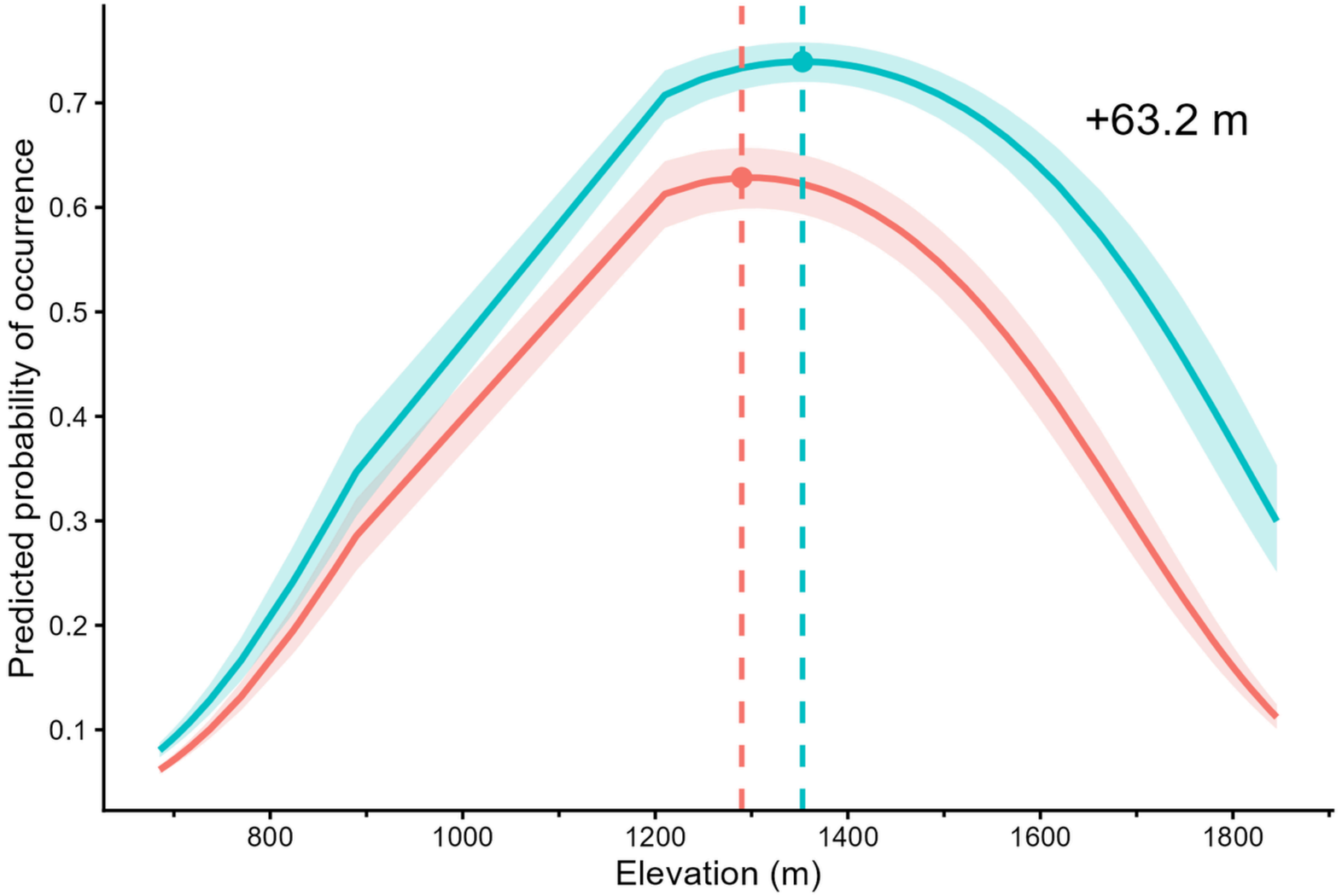
2. **Prediction:**
    - Increased occurrence of species at higher elevations in 2025 resurvey → upslope shift in peak elevation

Range shift results

Modelled the predicted probability of occurrence across elevation

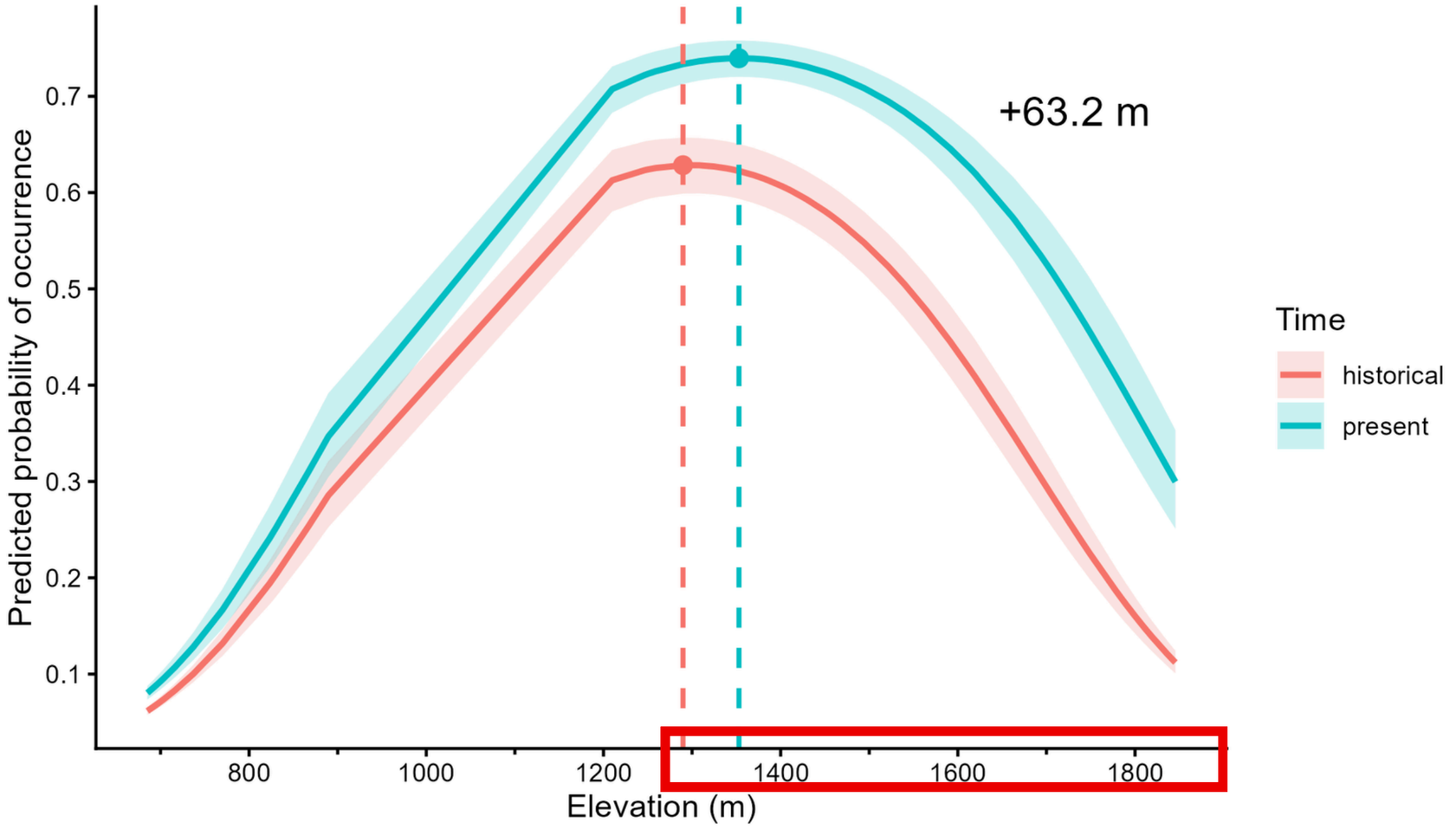


Significant upslope range shift in the optima of occurrence detected across species



- Significant interaction between time and linear elevation (β = 0.352, **p = 0.016**)
- Rate of **+12.7 - 24.9** m per decade
- Consistent with the literature
  - +11 m (Chen et al., 2011)
  - +29 m (Lenoir et al., 2008)

Increased probability of occurence at high elevations but not low elevations



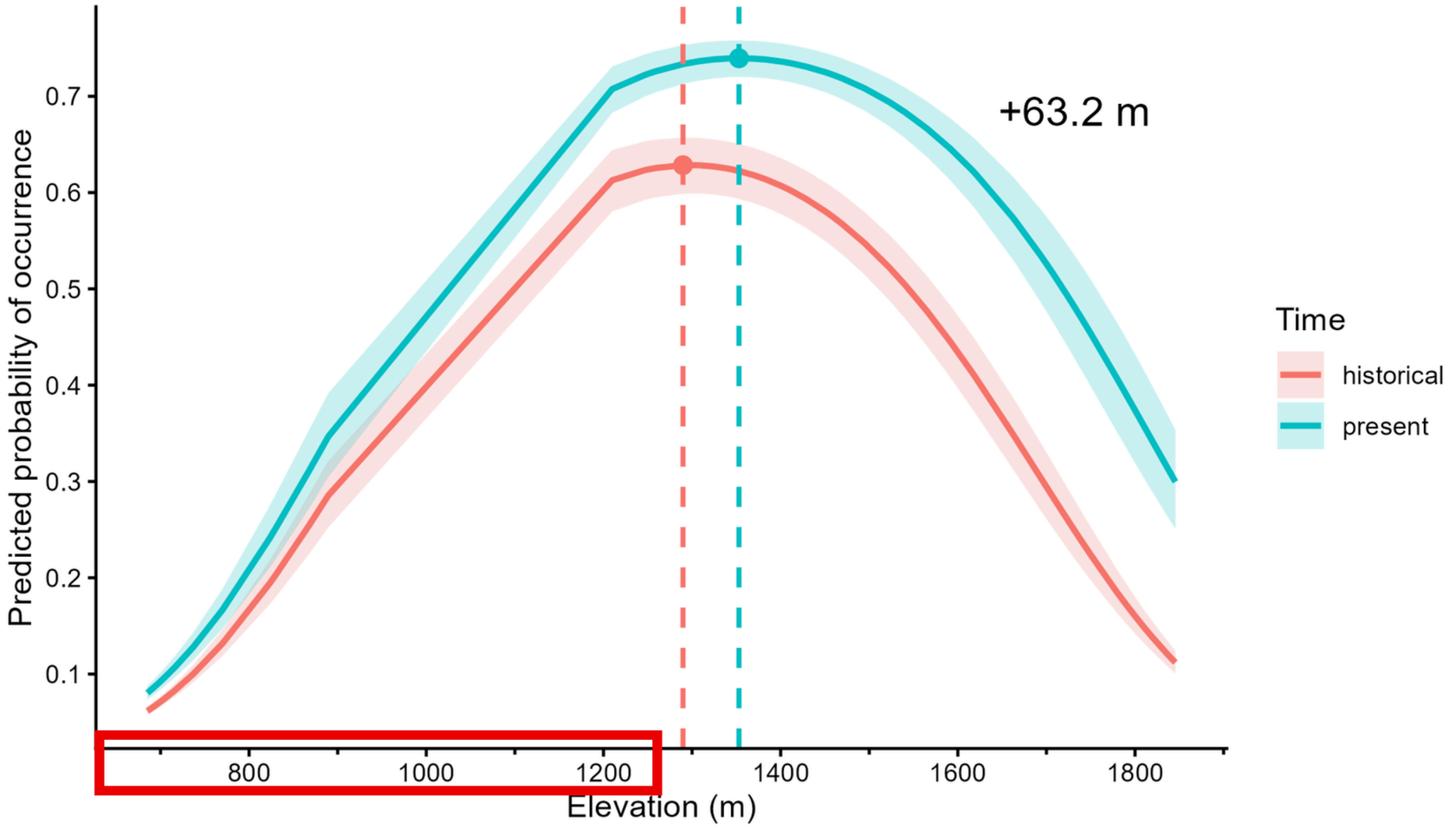
At high elevations/ leading edge:

- Species are being detected more frequently

In the literature:

- Increase of species richness on mountain summits (Steinbauer et al., 2018; Pauli et al., 2012)
- Strong signal of changing range dynamics, consistent with other montane systems

Increased probability of occurence at high elevations but not low elevations



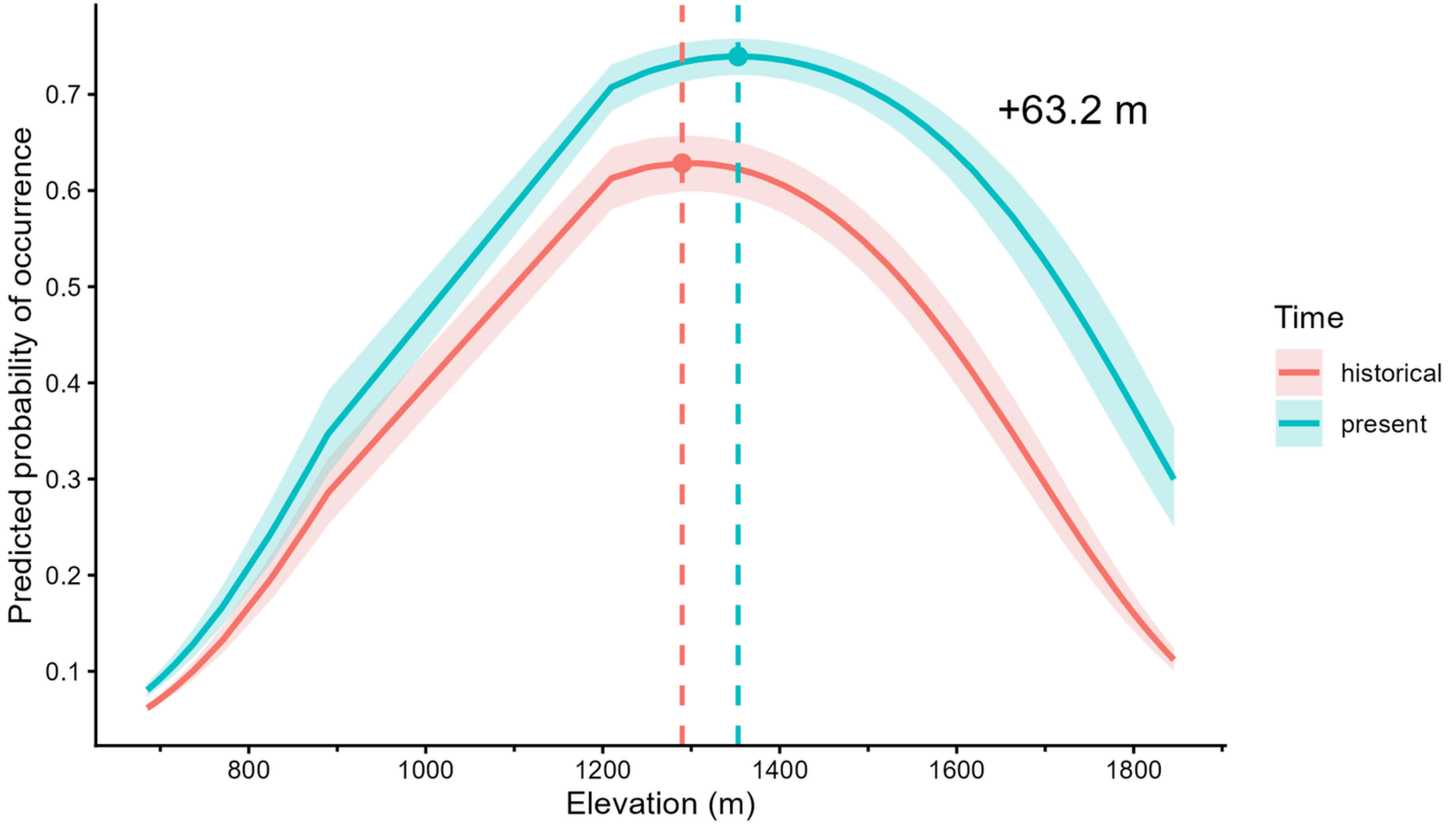
At low elevations/ trailing edge:

- No change from historical baseline
- Could suggest species are lagging behind climate

In the literature:

- Expansion occurs before local extinctions are realized (Alexander et al., 2018)
- Microclimate buffering (De Frenne et al., 2013)

Hard to break down the range shift signal

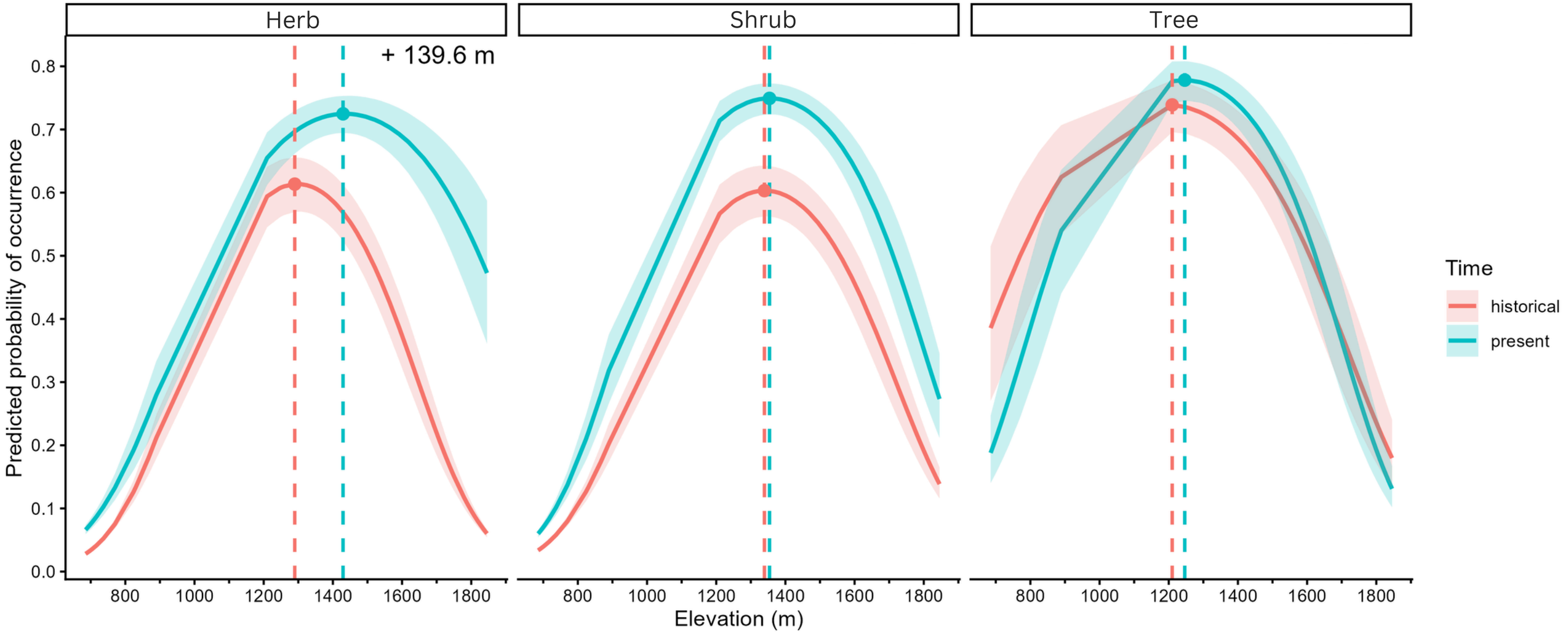


The range shift is detected across 30 species and by the optima of occurrence

Signal could be due to or hiding:

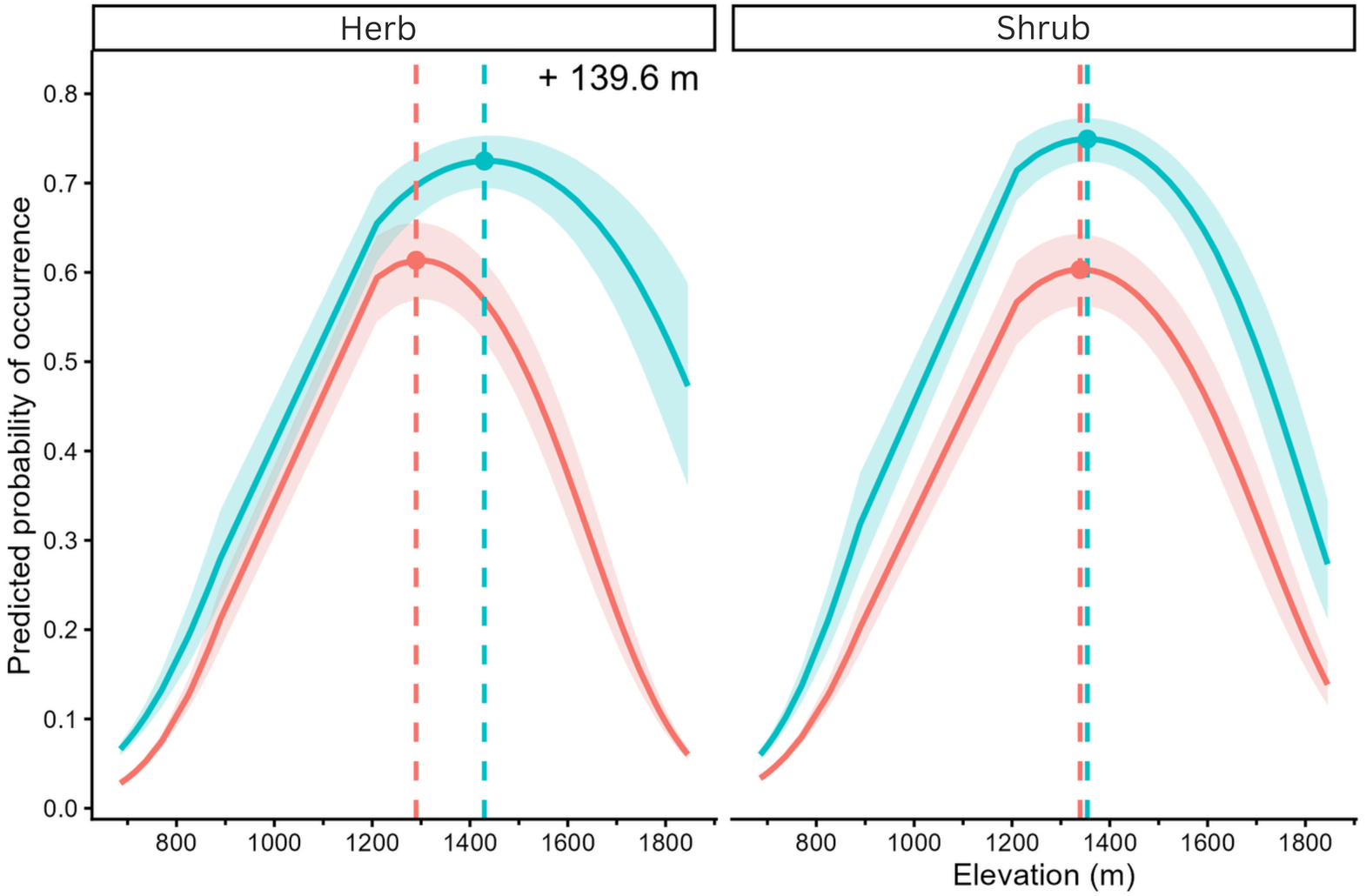
- Range expansion or contraction
- Shifts at leading and/ or trailing edges
- Increased abundance and detection of species at high elevations

Functional groups exhibit vastly different responses across elevation



- Significant upslope shift (β = 0.805, p = 0.0019)
- Uniform increase in probability of occurence (β = 0.674, p = 0.0056)
- No significant change

Herbs may be tracking climate better than shrubs and trees



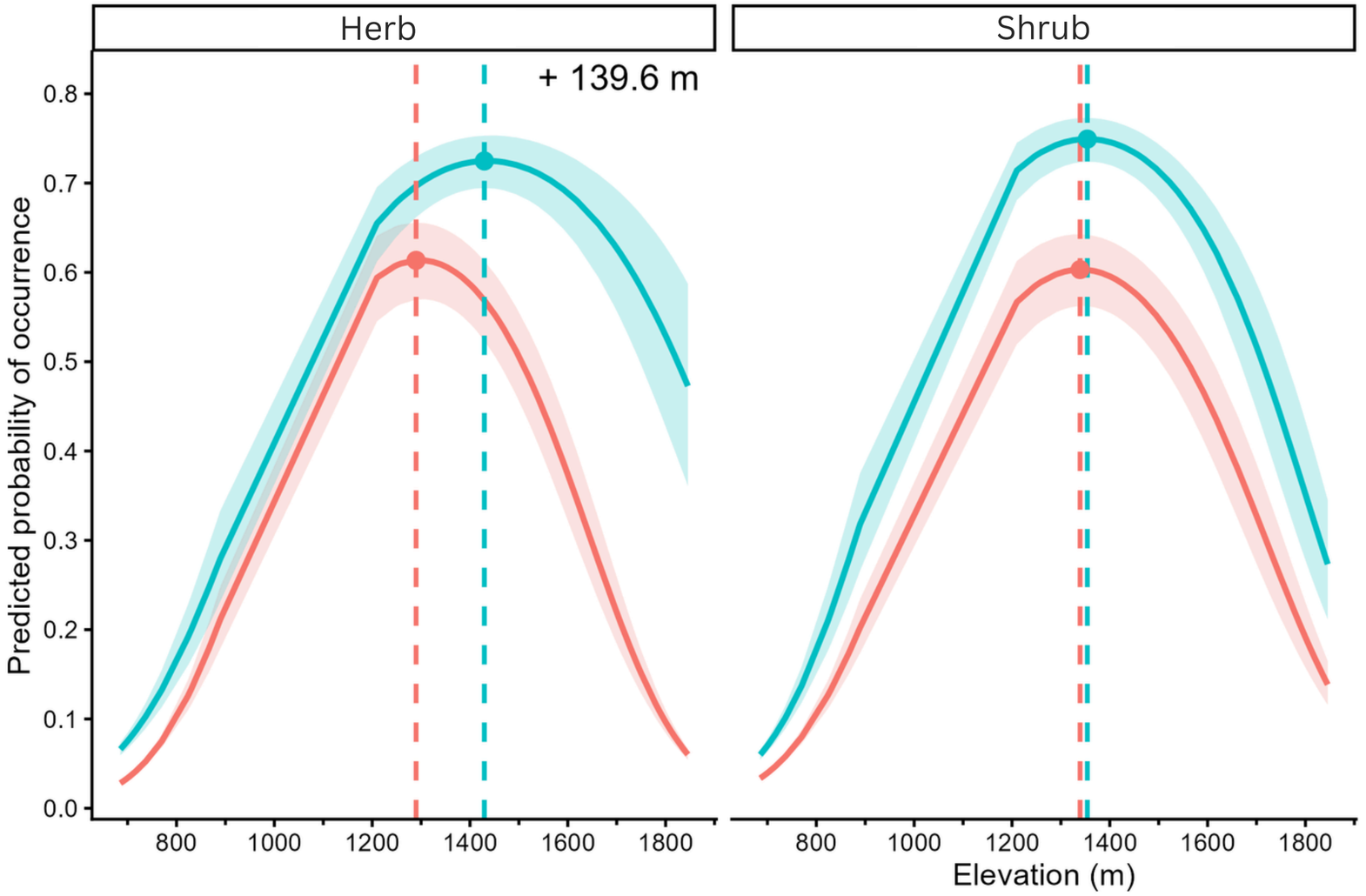
Herbs:

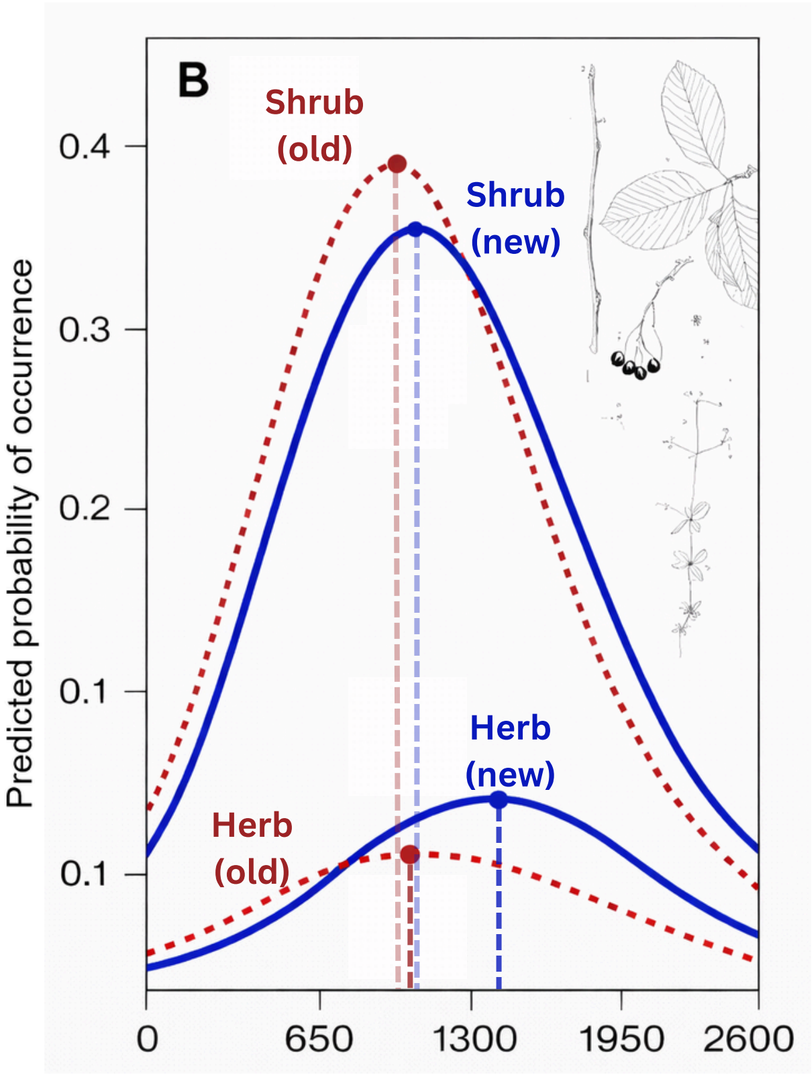
- Non-woody, shorter-lived, faster-turnover taxa may track climate better

Shrubs and trees:

- Woody species with slower demographic turnover may be lagging behind climate

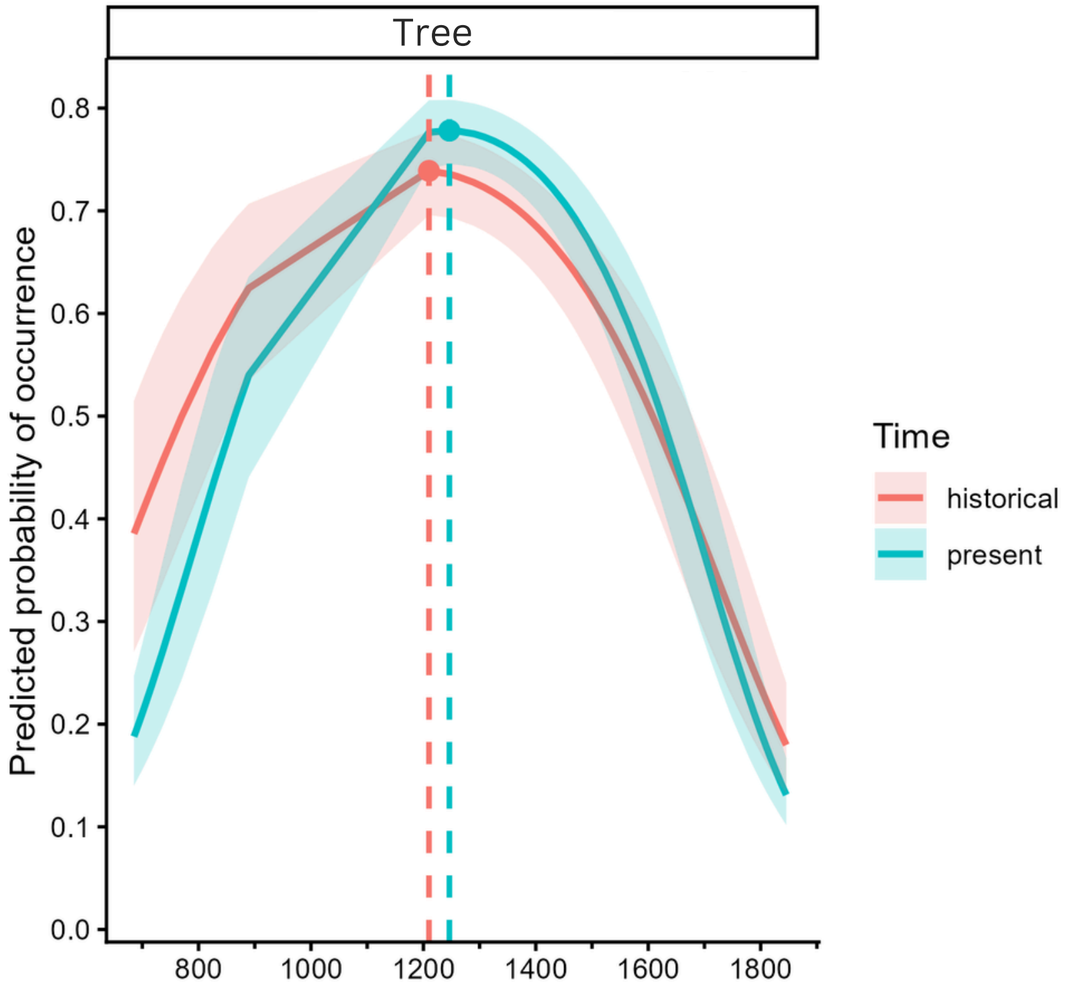
Herbs may be tracking climate better than shrubs and trees





Source: Lenoir et al. (2008)

Trees may be not be tracking climate fast enough



No change across elevation:

- Could suggest trees are lagging behind climate

In the literature:

- Current distributions are underfilling their full climatically suitable ranges (Seliger et al., 2021)

Future direction:

- Use extensive tree height data to track changes in stand composition
- Look to detect an increase of shorter, younger trees at high elevations

Summaries & future directions

Summary and future directions of relocation analysis

What I found:

1. Relocation error did not strongly influence results across any diversity metrics tested

2. Change between 2019 and 2025 may likely be influenced by the 2021 North American Heat dome

Future directions:

- BEC data present a rare and exciting opportunity to resurvey diverse ecosystems and environmental gradients in British Columbia

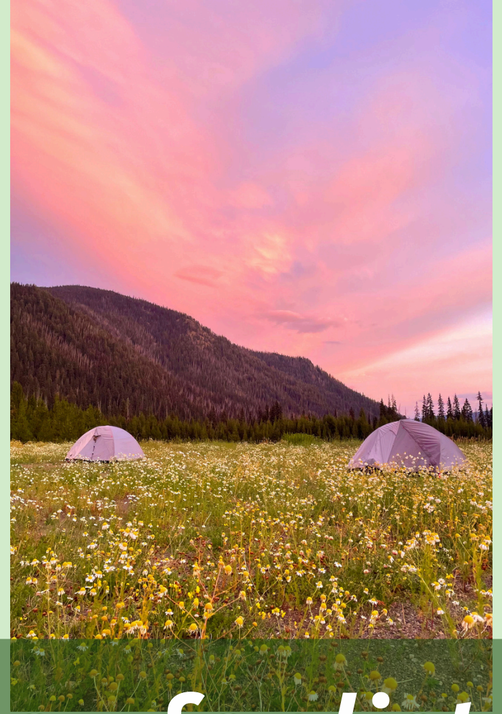
Summary and future directions of vegetation change analysis

What I found:

1. Significant upslope range shift across species
2. Increased probability of occurence at high elevations but not low elevations
3. Herbs may be tracking climate better than shrubs and trees

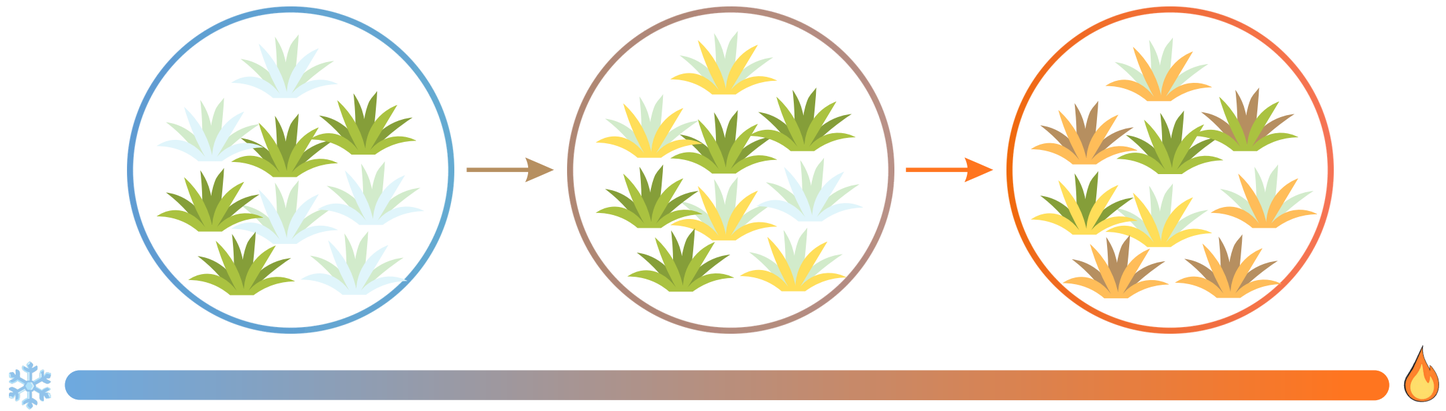
Future directions:

- Investigate how community composition is changing
  - Use my data to test for thermophilization across an elevational gradient
- Investigate changes in tree stand composition



Thank you for listening!

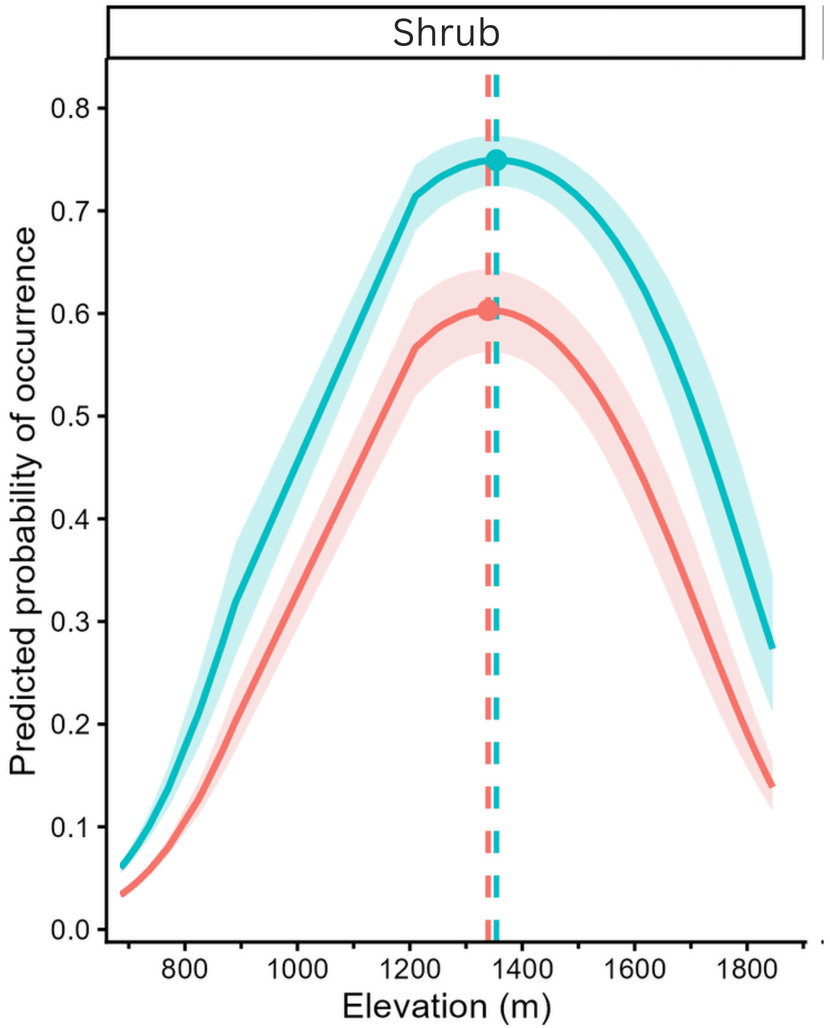
Future direction: test for community thermophilization



Thermophilization

- The process where plant communities are becoming increasingly dominated by warm-adapted species

Early signs of shrub encroachment



Uniform increase in the probability of occurrence:

- Could suggest shrubification or shrub encroachment

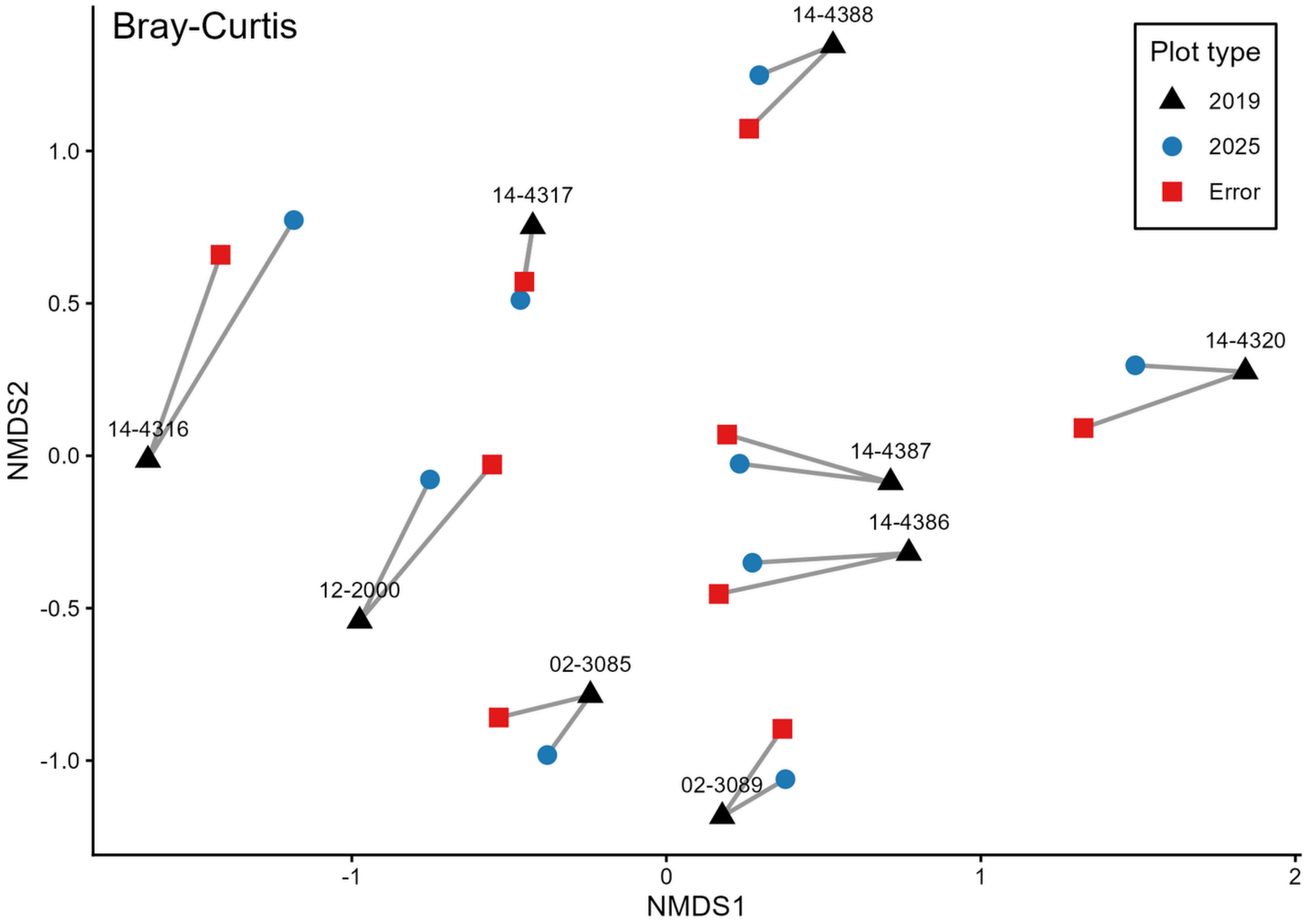
In the literature:

- Evidence for shrub encroachment in the subalpine (Zeidler & Banaš, 2024)

Impacts:

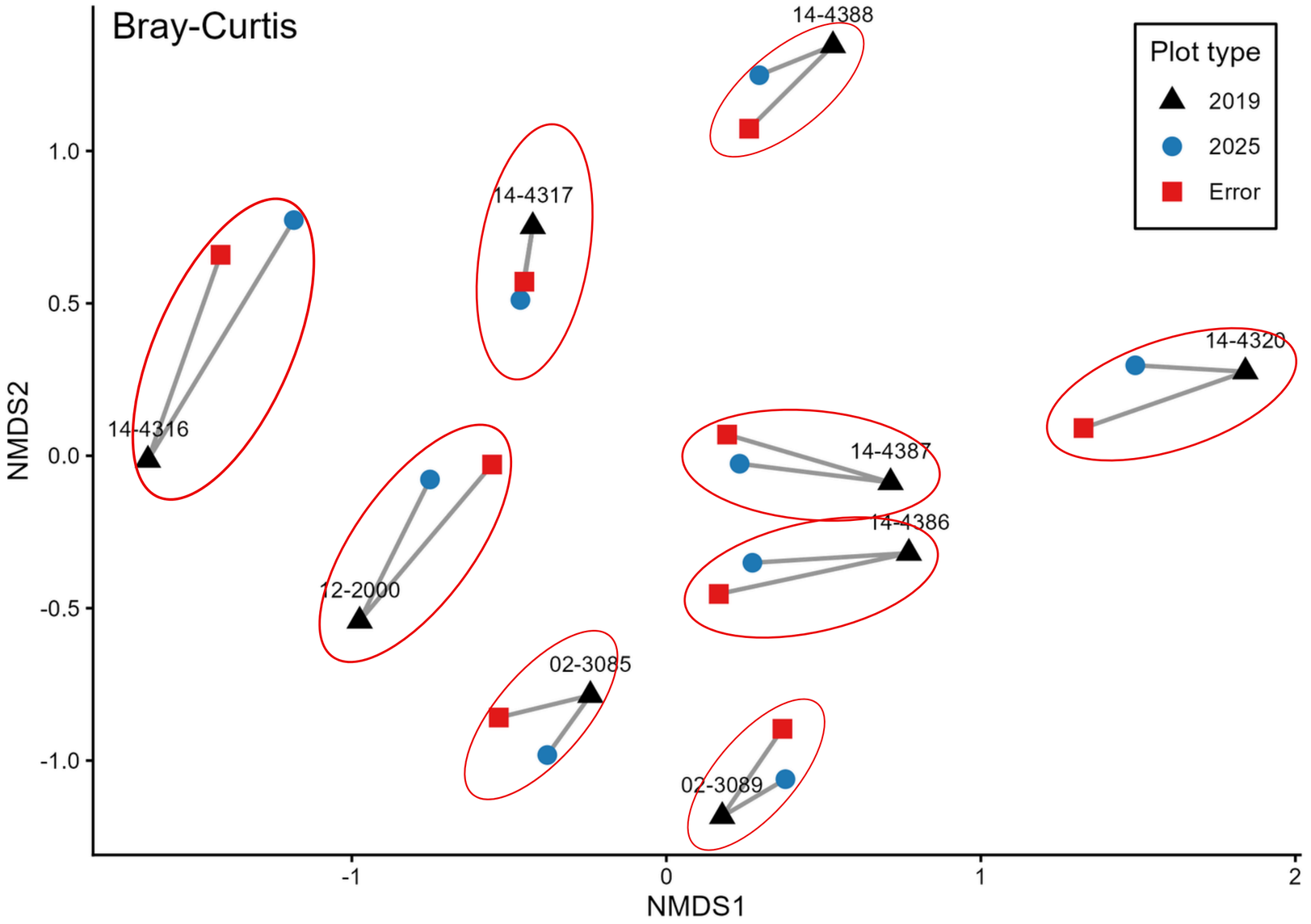
- Both positive (by maximizing diversity at moderate abundances) or negative (reducing herb diversity)

Significant difference between 2019 and 2025 plots



- 2019 vs. 2025$_{\text{Correct}}$ (**p = 0.0039 ****)
- 2019 vs. 2025$_{\text{Error}}$ (**p = 0.0078 ****)

Significant difference between 2019 and 2025 plots



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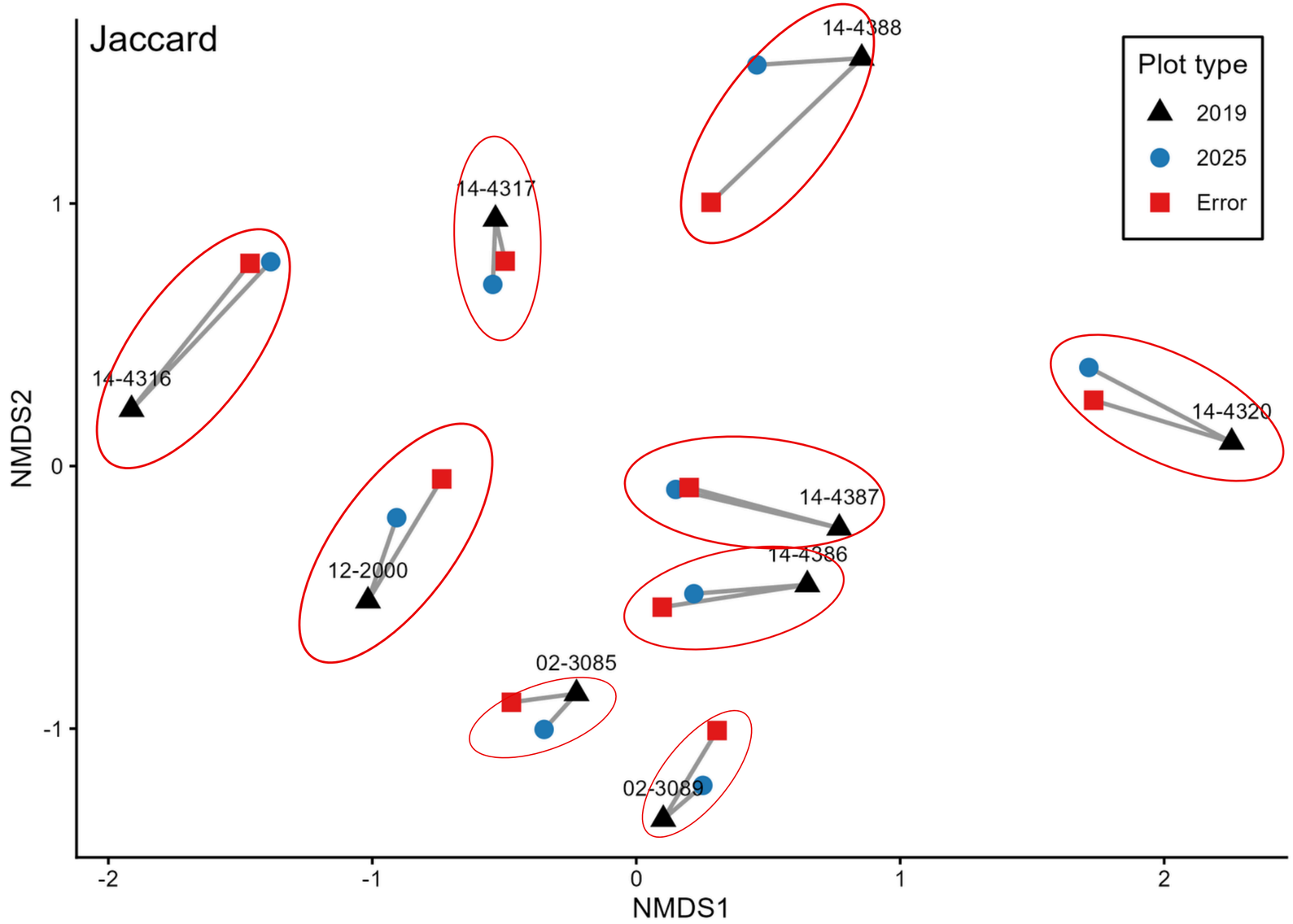
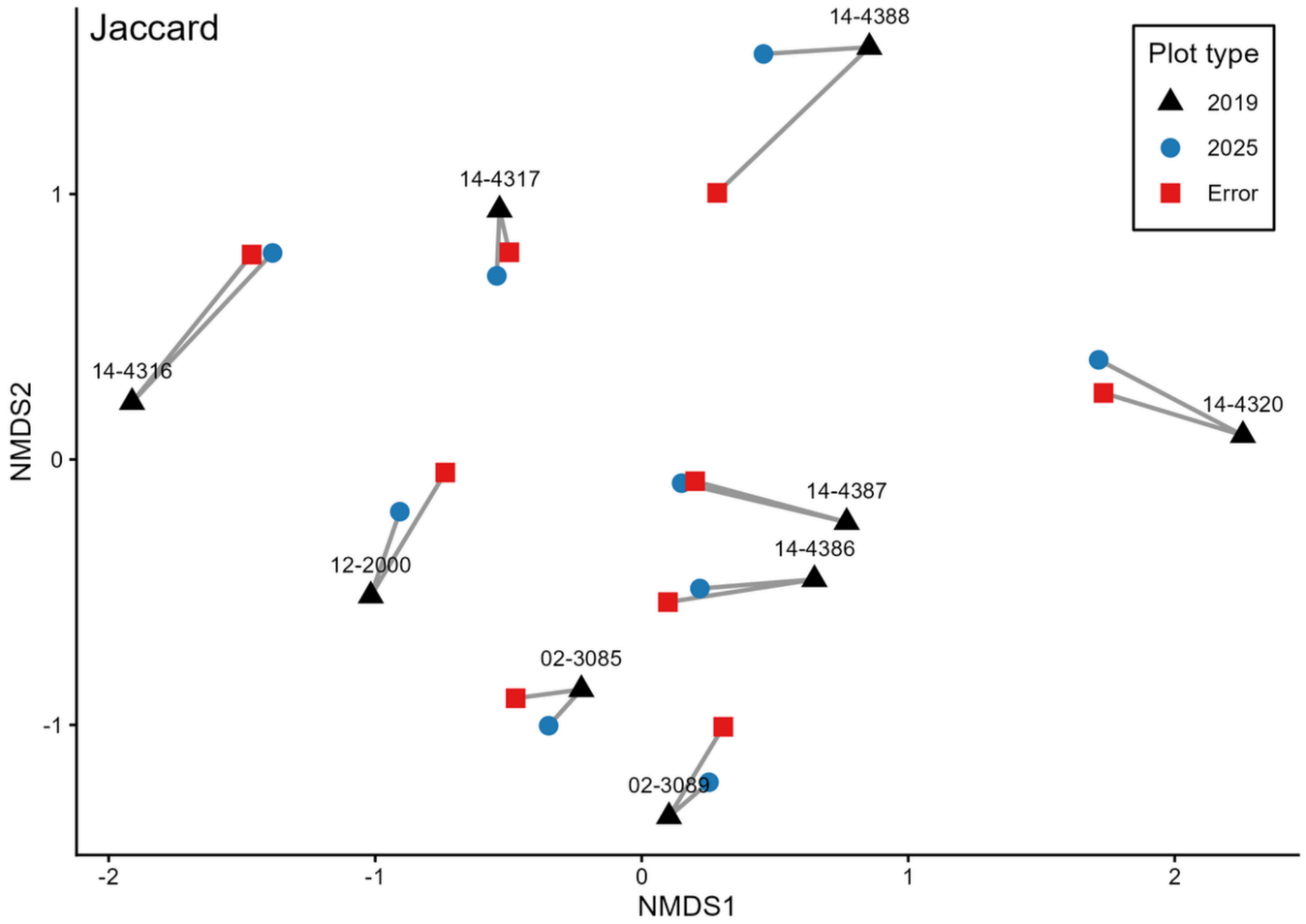
Significant difference between 2019 and 2025 plots



- 2019 vs. 2025$_{Correct}$ (**p = 0.0078 ****)
- 2019 vs. 2025$_{Error}$ (**p = 0.0039 ****)

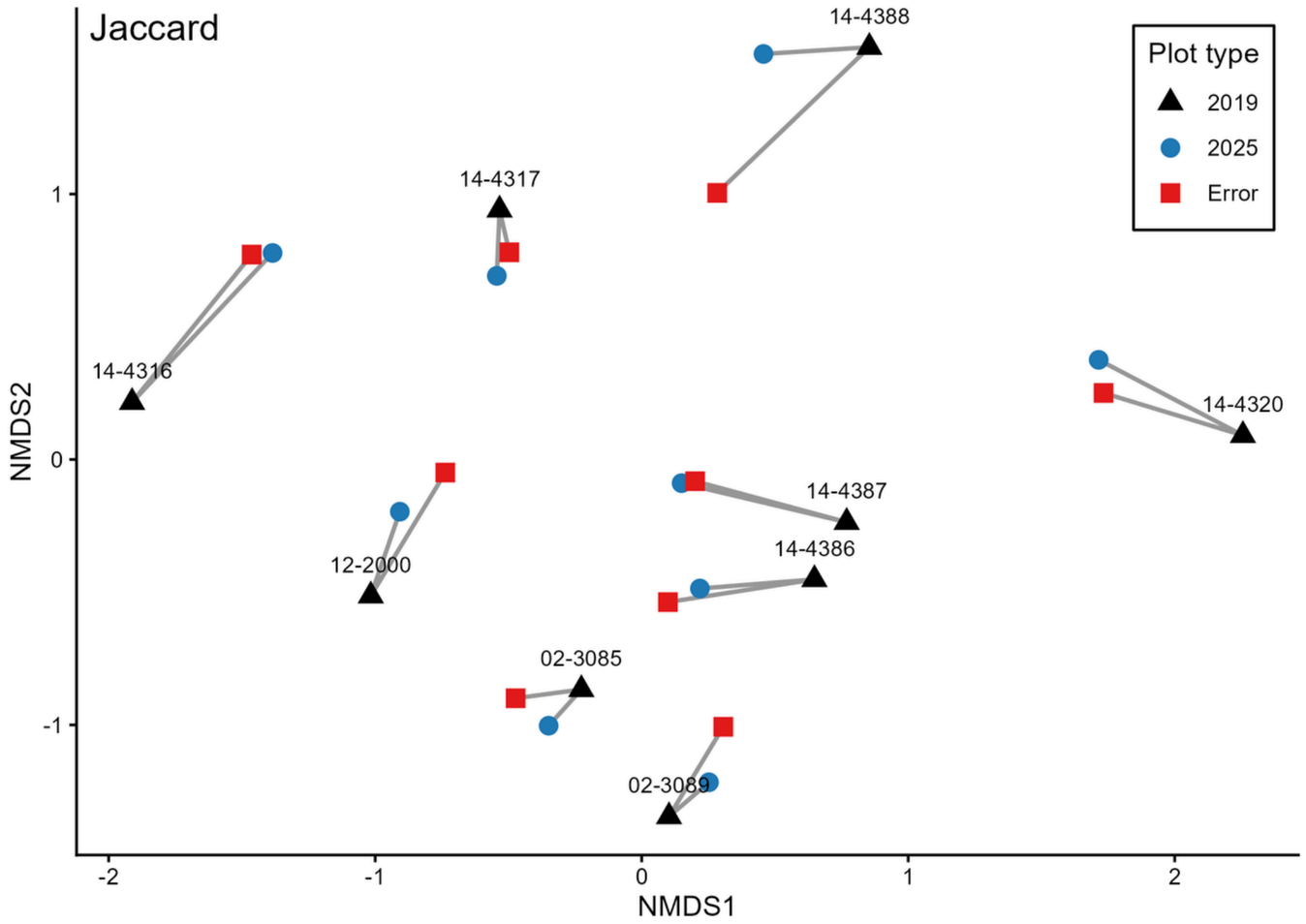
Significant difference between 2019 and 2025 plots



2 hypotheses:

1. Artifact of surveyor bias
2. 2021 North American heat dome caused detectable compositional change

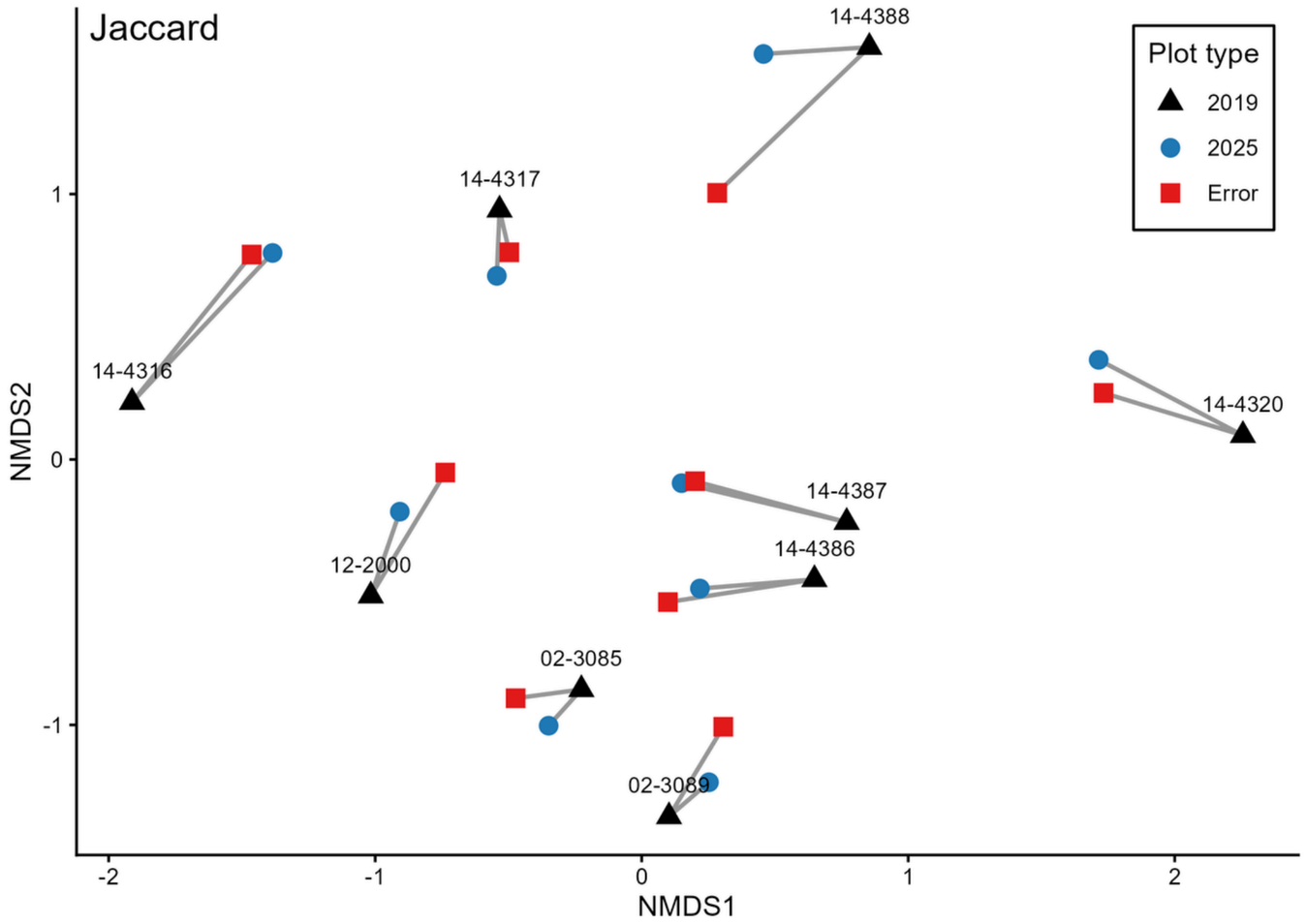
Surveyor bias?



Reasoning for:

- Differences in species identification or cover estimation between survey teams
- Could create pseudo-turnover
  - Especially likely for difficult taxa (e.g., Vaccinium spp.)

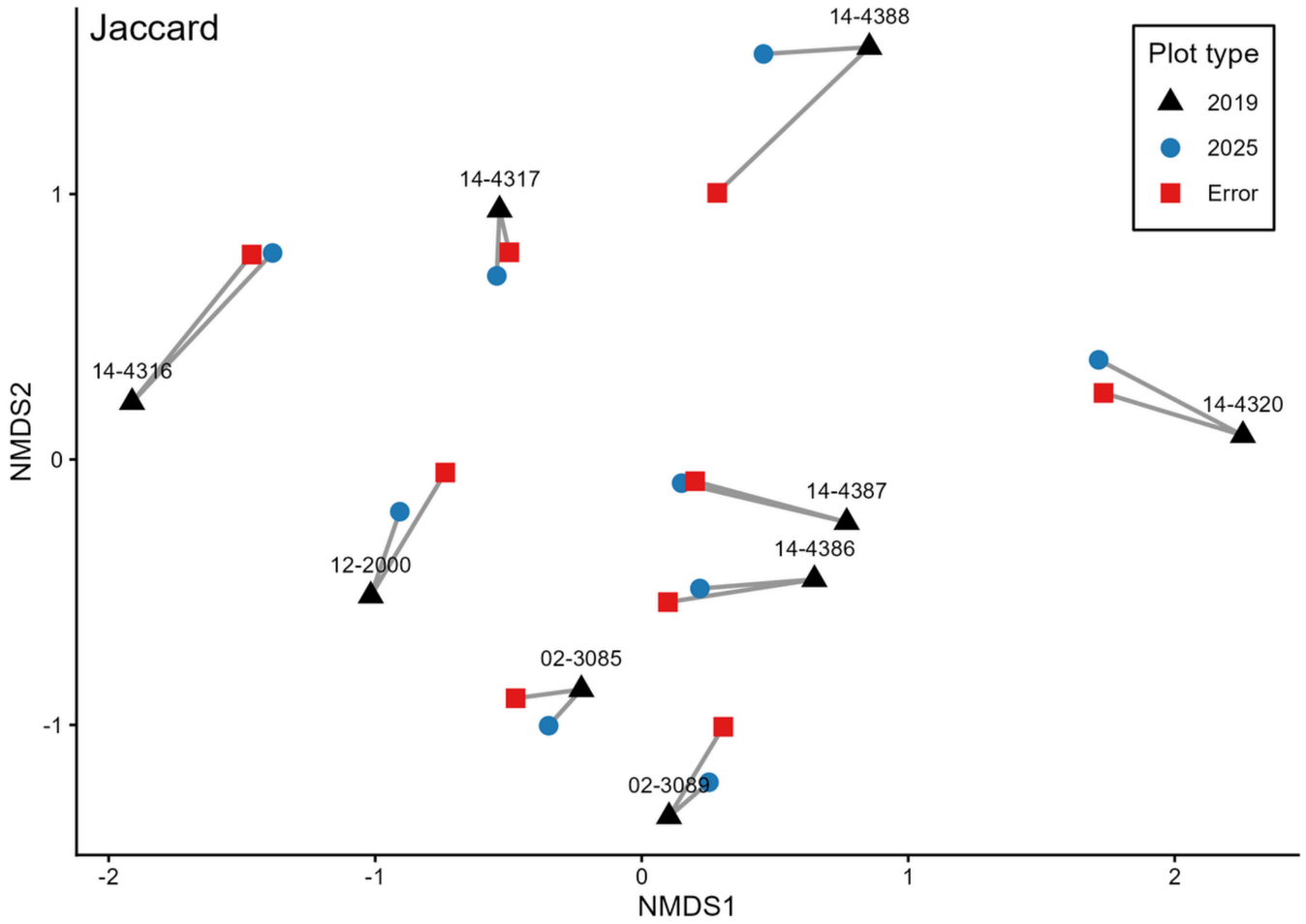
Surveyor bias?



Reasoning against:

- Consistent results across Bray-Curtis & Jaccard
- No differences in richness, Shannon, Simpson
- Same BEC protocols + trained surveyors

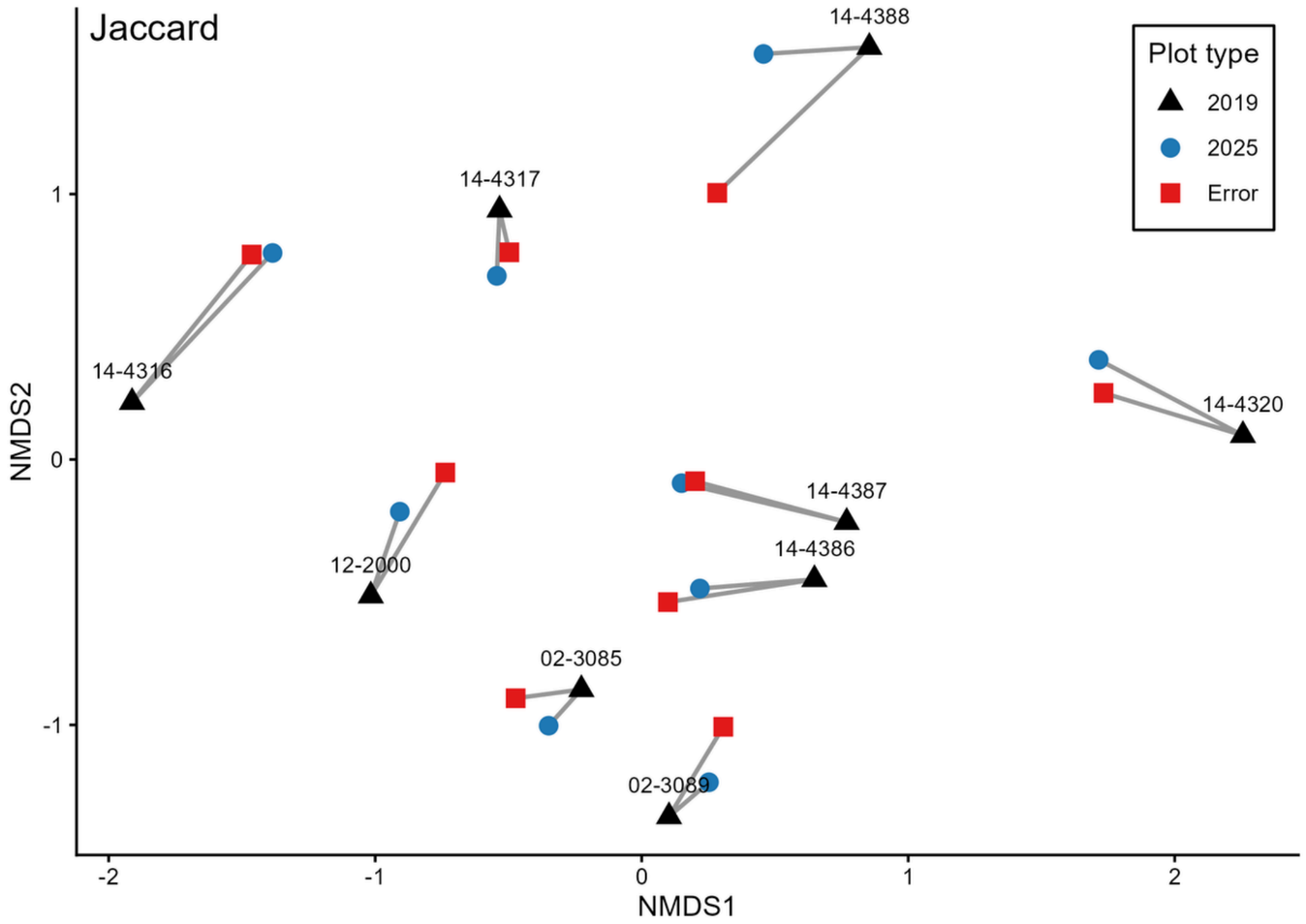
2021 North American heat dome?



Extreme event:

- Extreme event: +21°C above average, ~37–42°C for 7 days
- One of the most extreme events ever recorded (Thompson et al., 2022)
- Baum et al. (2026) found **75%** of species in the region were **negatively impacted**
  - Understory herbs and shrubs

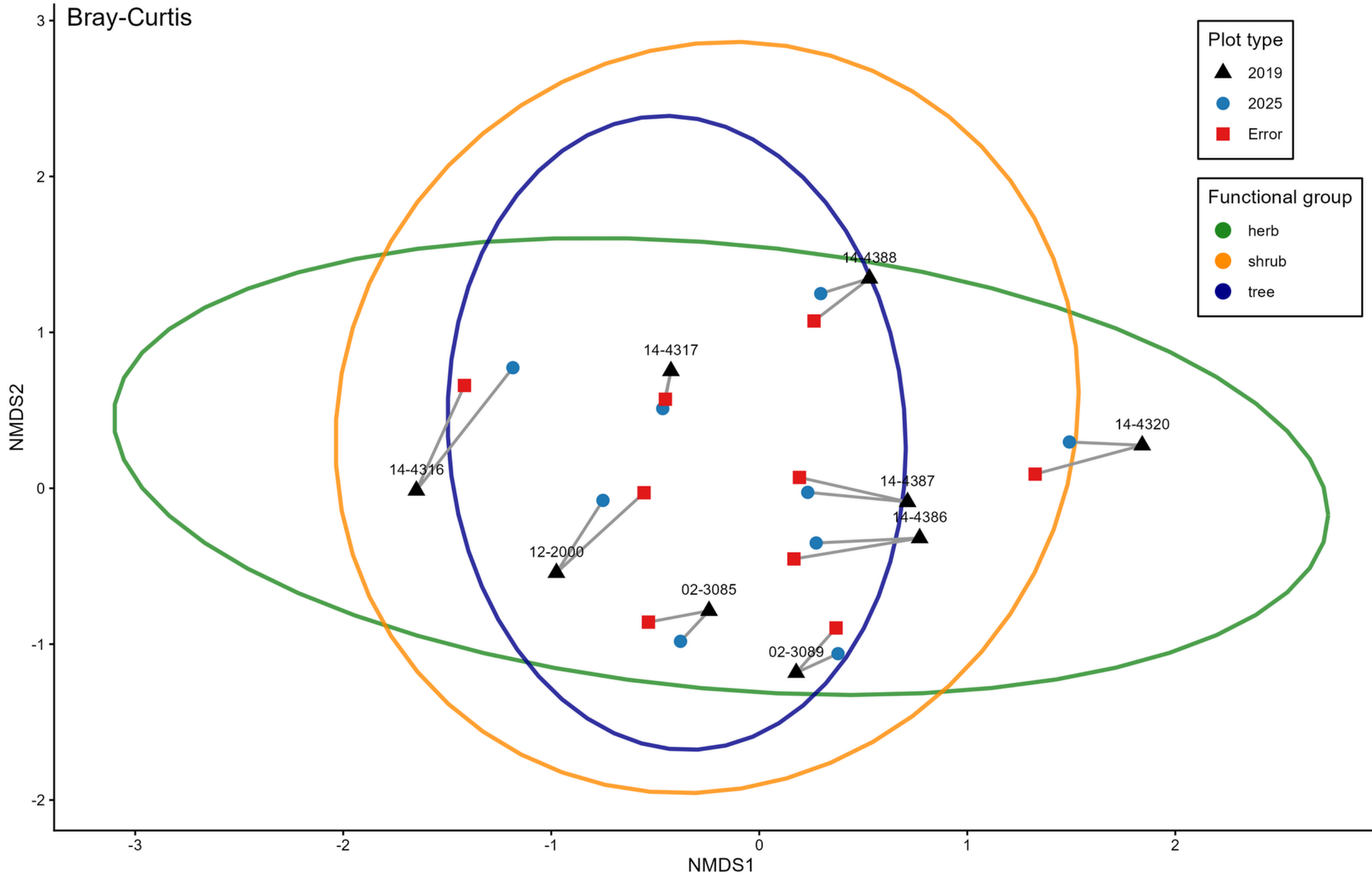
2021 North American heat dome?



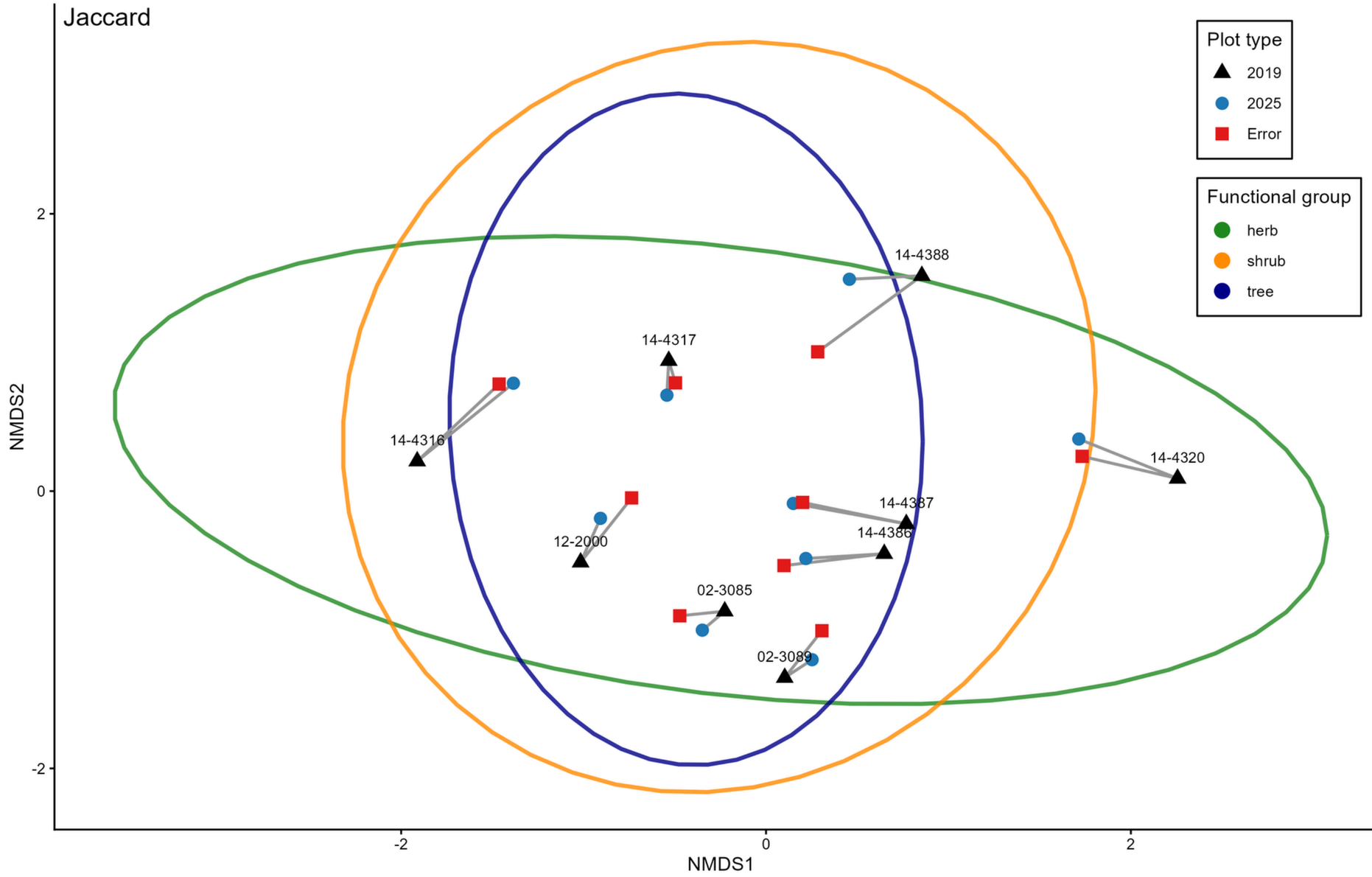
Highlights:

- Extreme heat events like the 2021 heat dome may drive rapid, vegetation change over short timescales
- Important to incorporate these disturbances into long-term monitoring frameworks

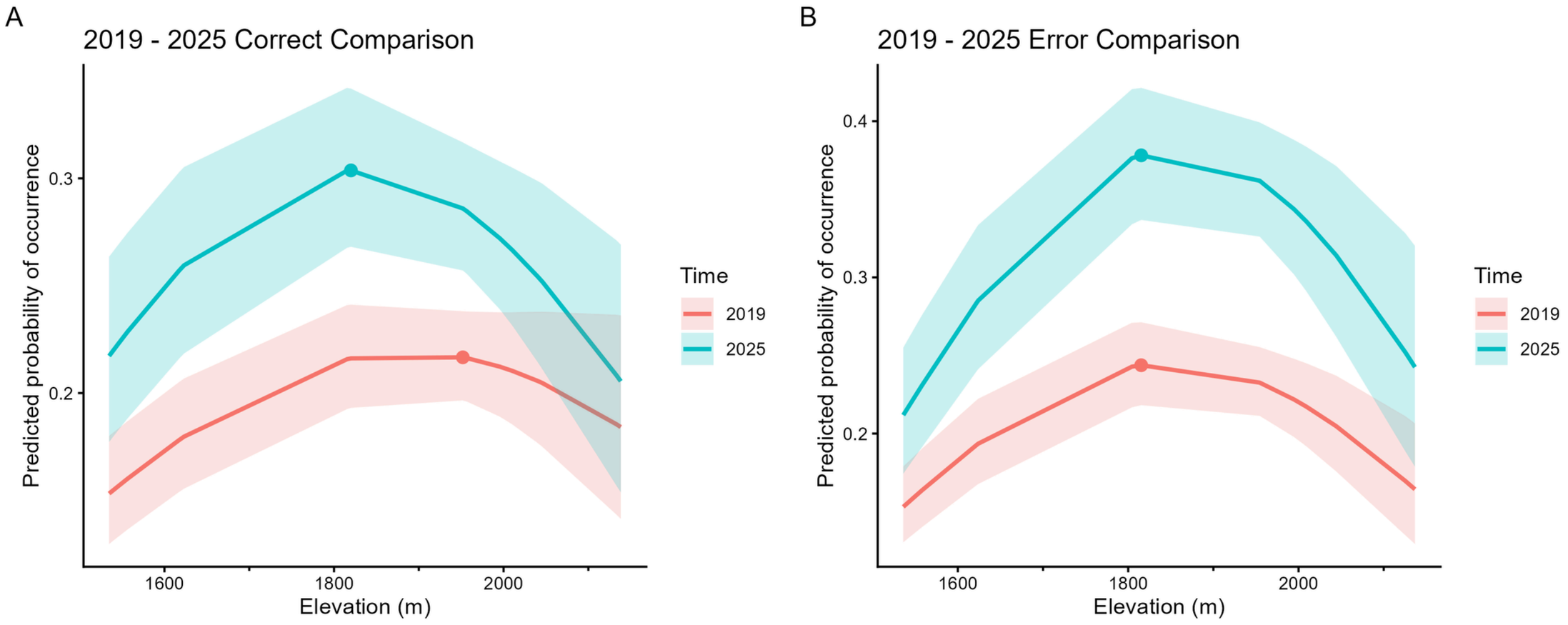
Functional group 95% confidence elipses



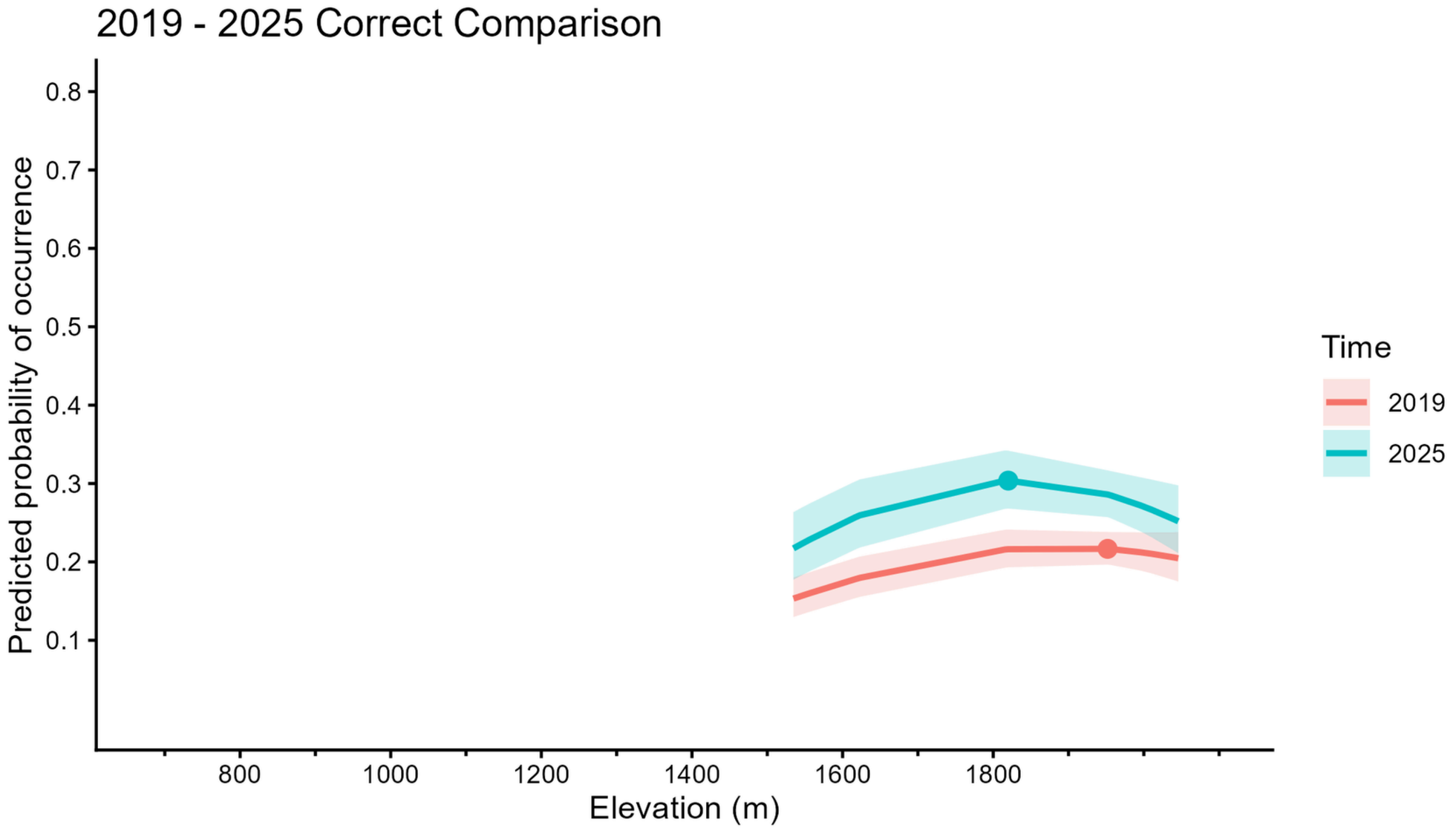
Functional group 95% confidence elipses



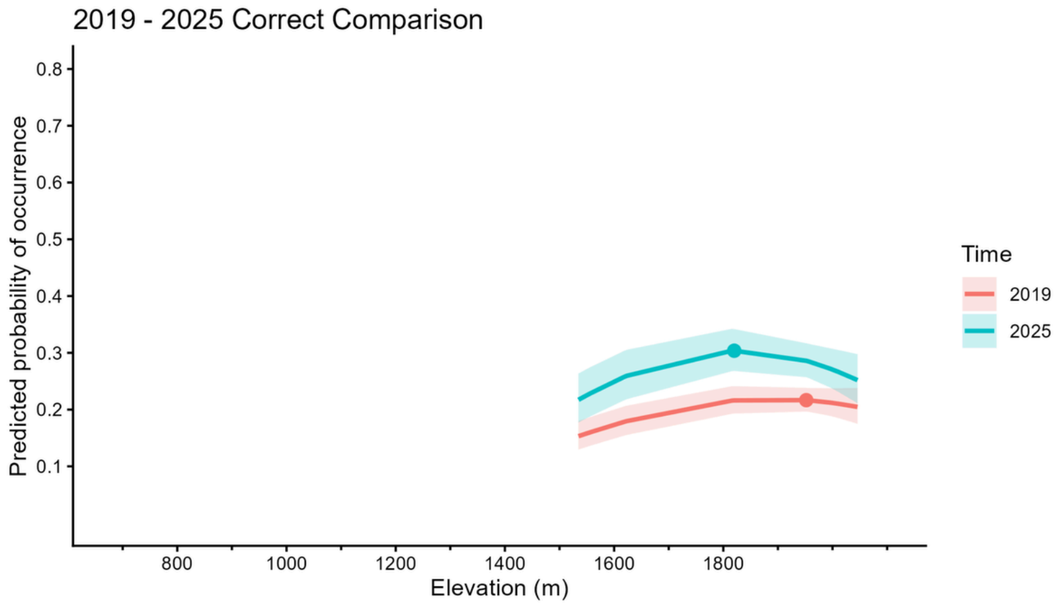
Elevation by time model on 2019 data

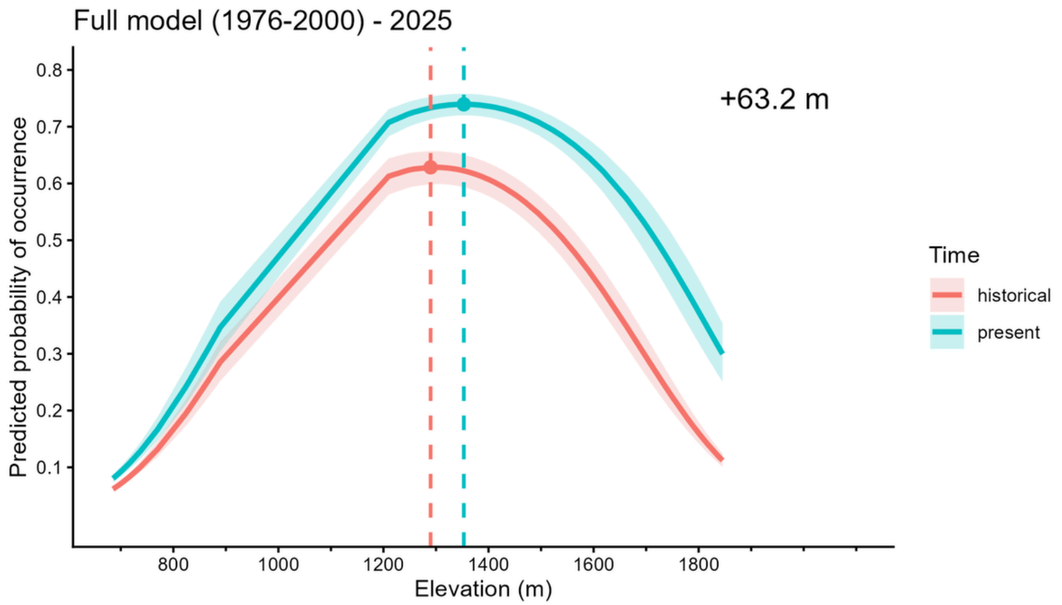


Elevation by time model on 2019 data

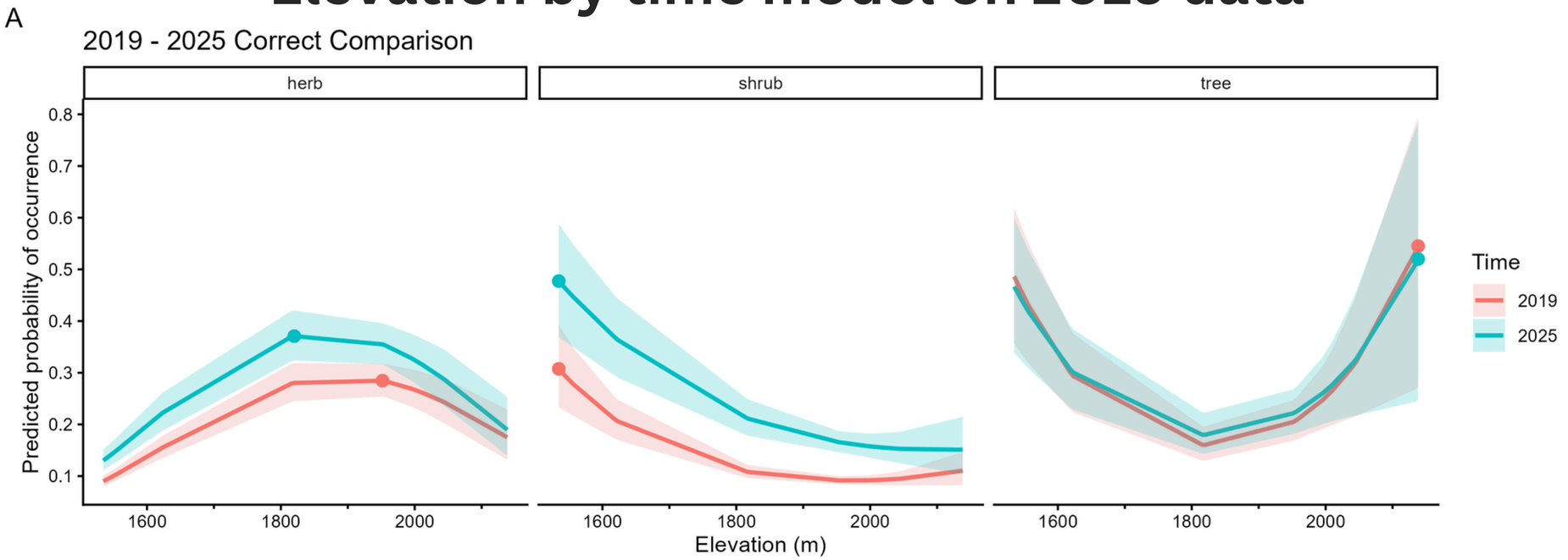


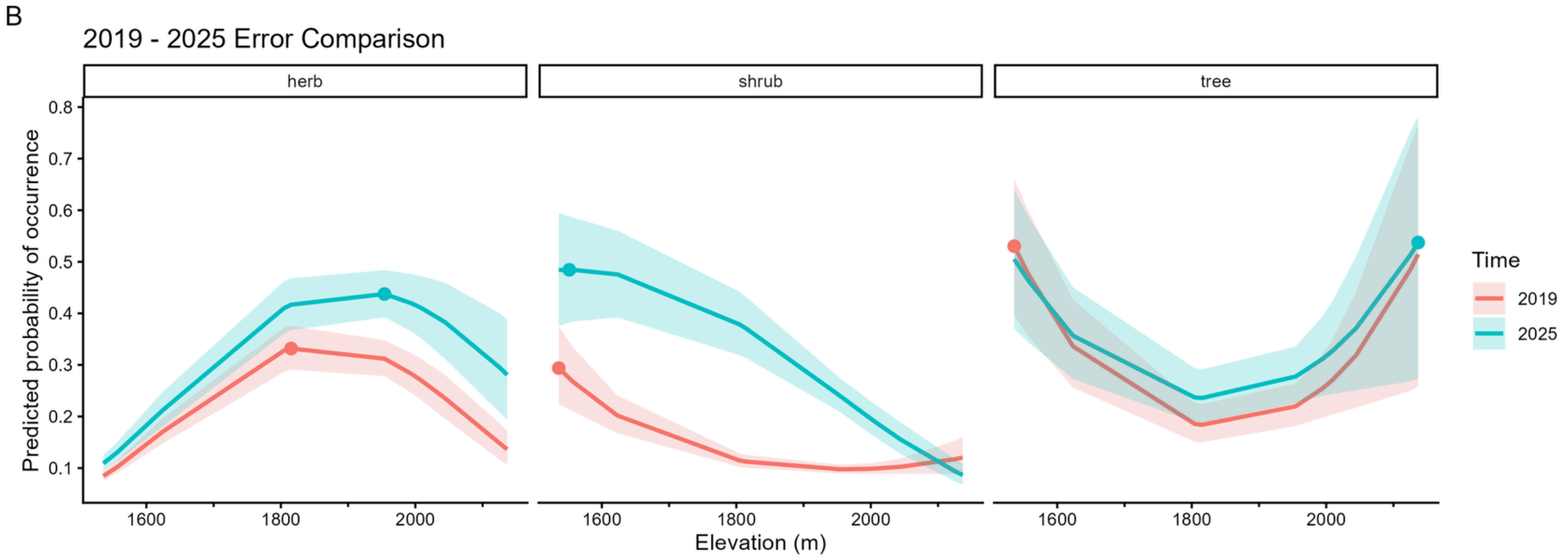
Elevation by time model on 2019 data



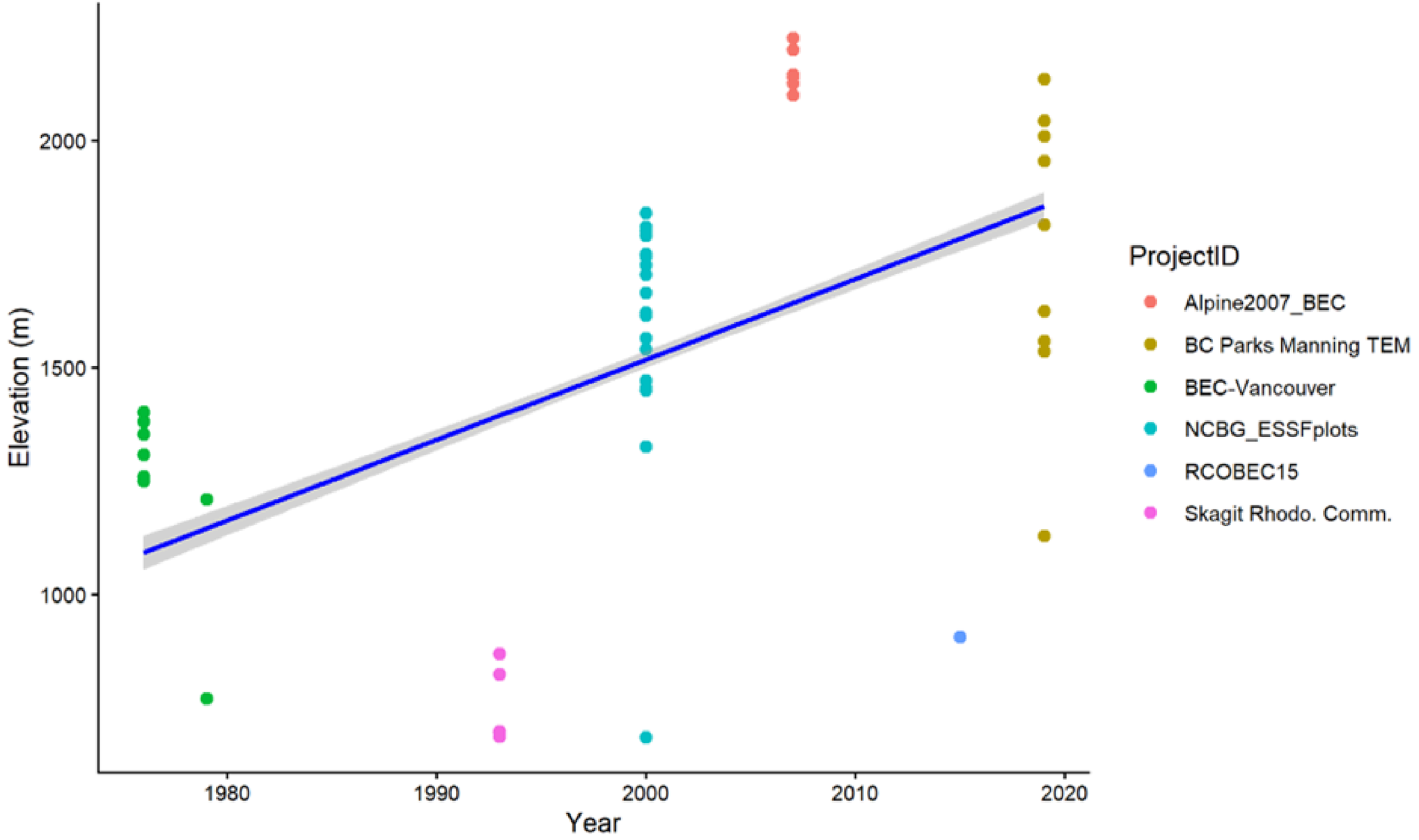


Elevation by time model on 2019 data

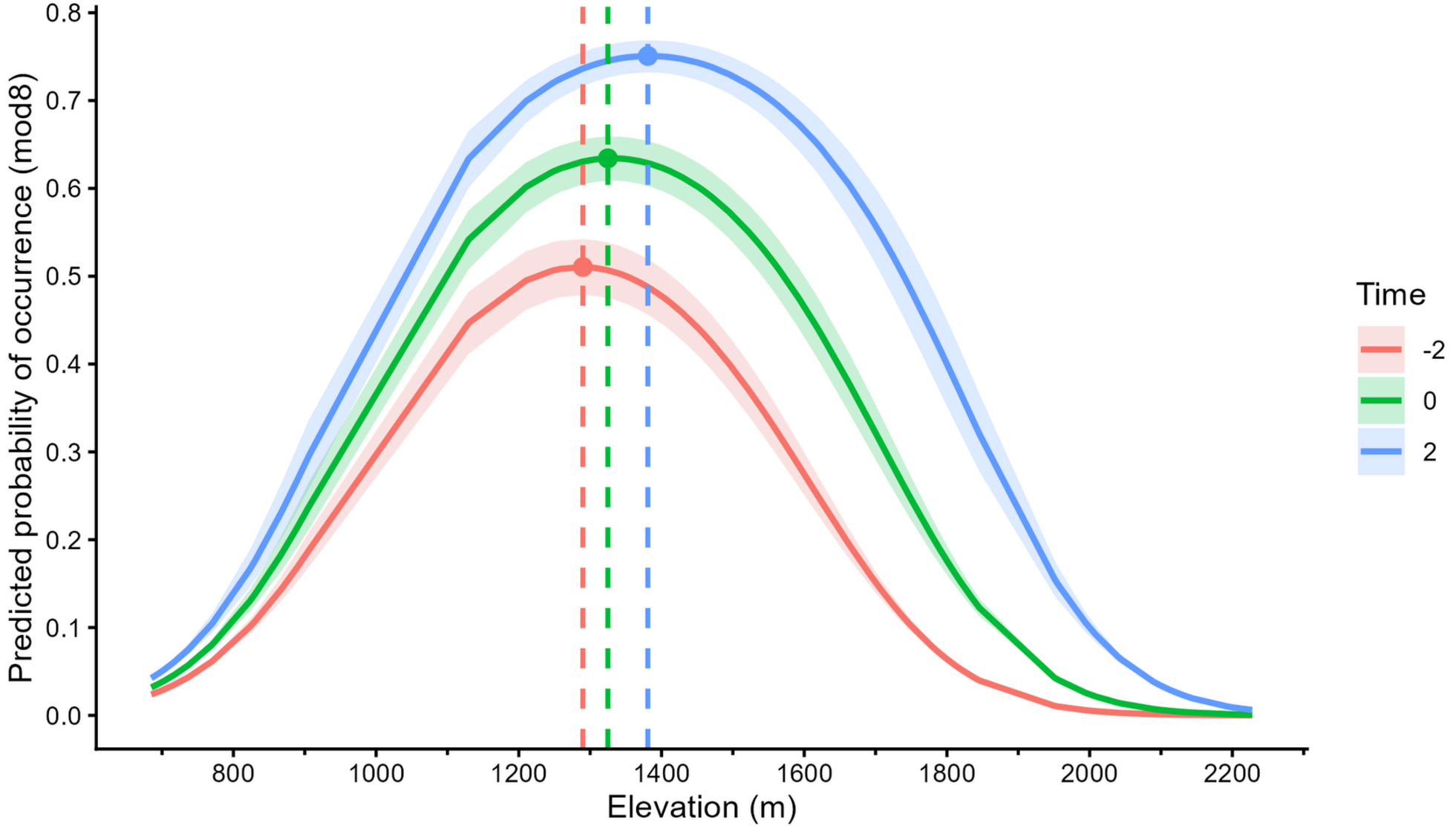




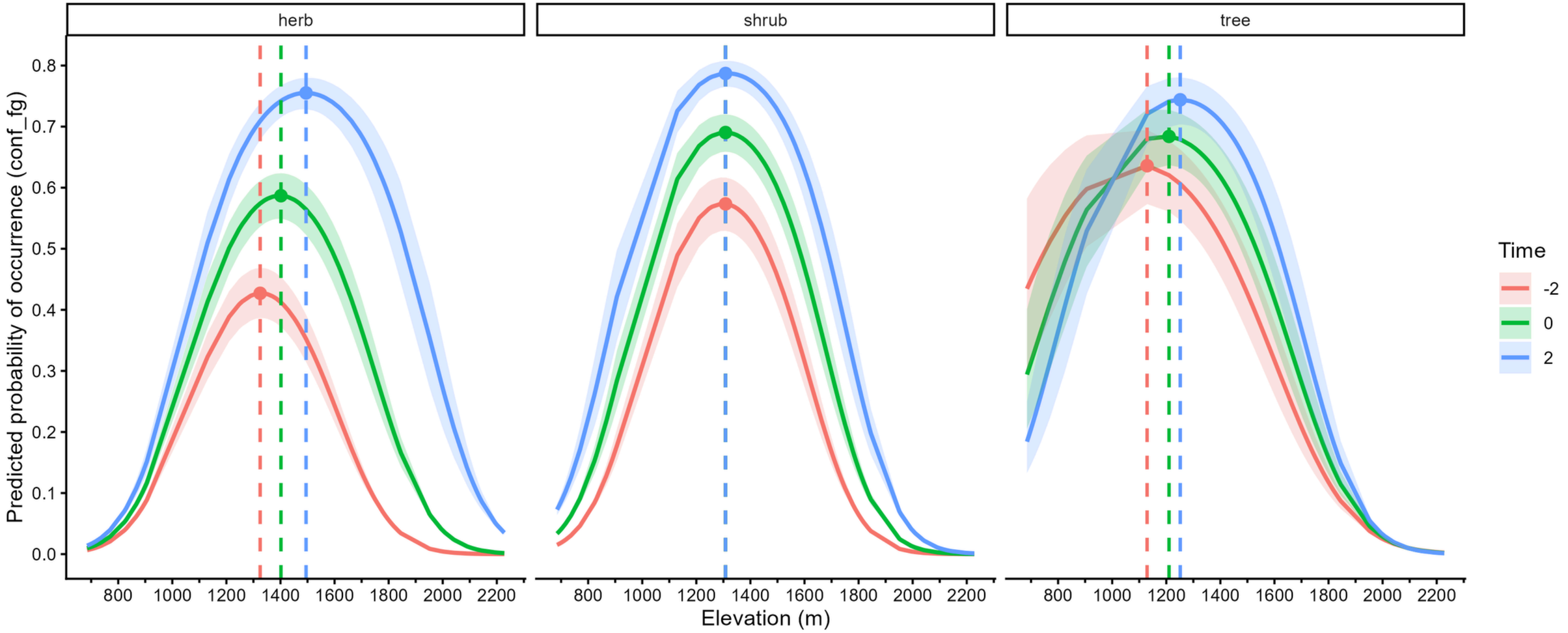
Significant correlation between survey year and elevation



Detecting upslope range shifts with continuous time

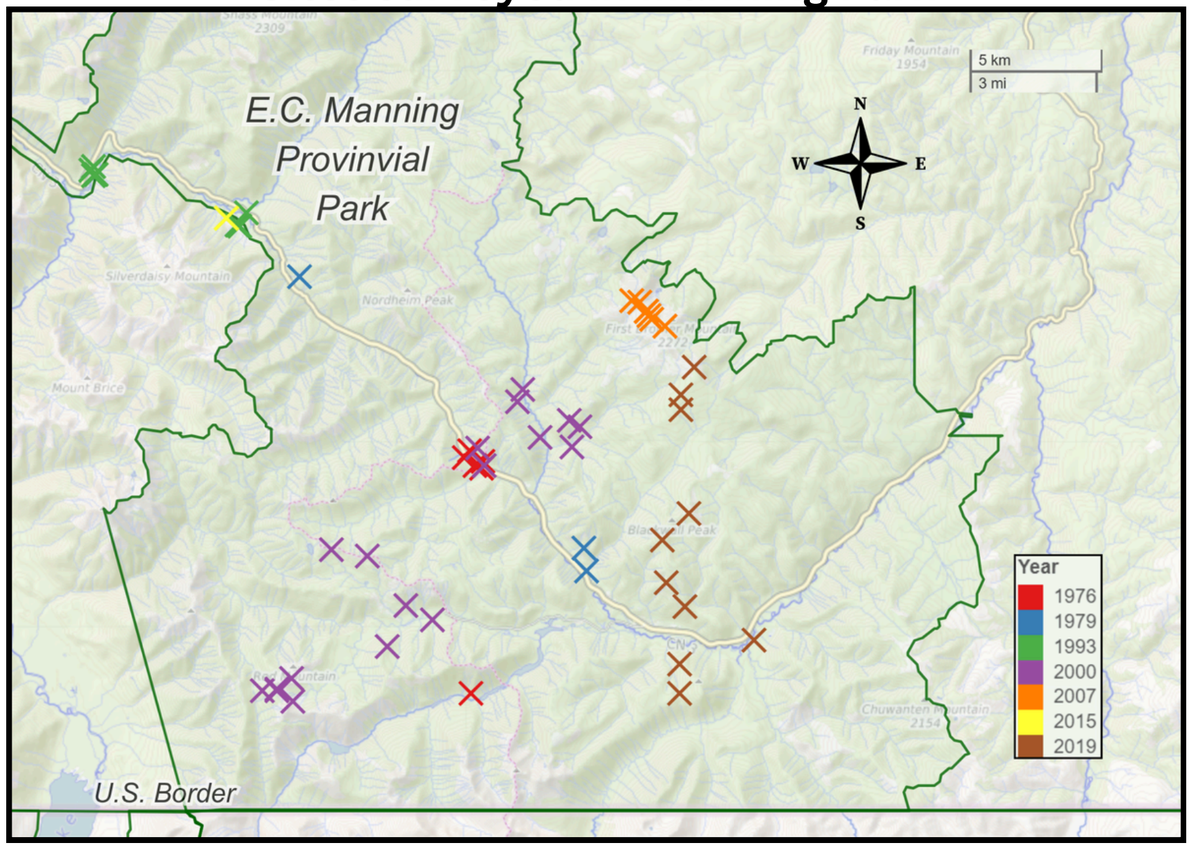


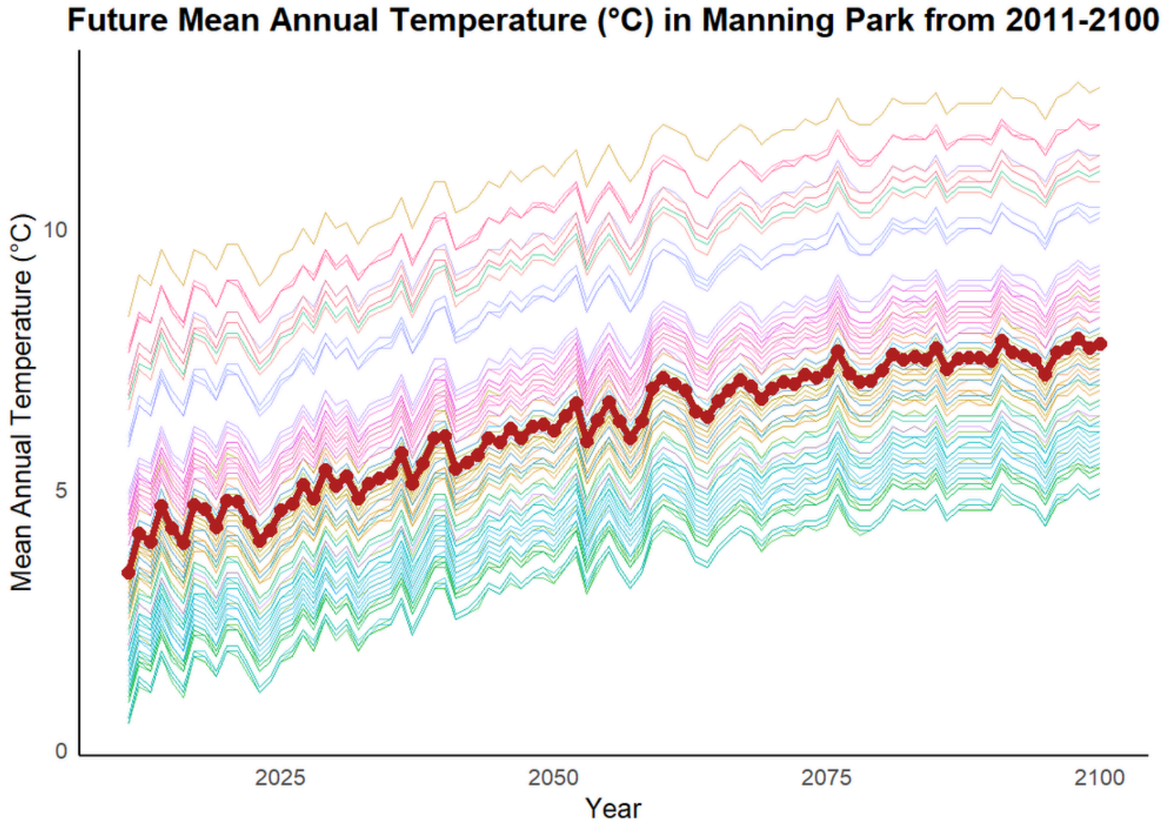
Detecting upslope range shifts with continuous time



Sites in Manning Park have experienced 1.54 °C in warming on average compared to the 1961-1990 historical normal

Sites surveyed in Manning Park





Source: OpenStreetMap, ClimateNA

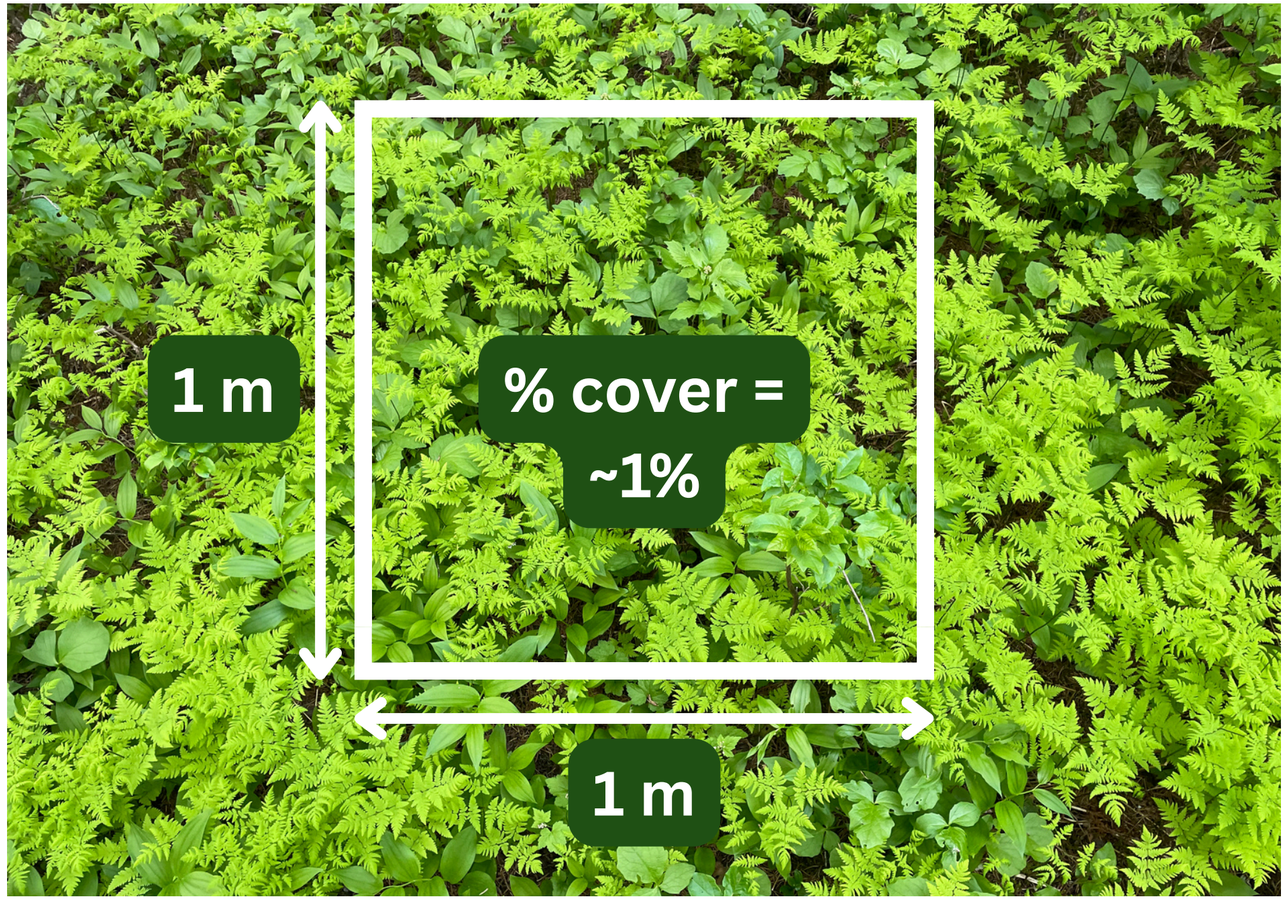
Observations (n)	Sites (n)	Species (n)
2040	34	30

Species selection criteria

- **≥ 20** total occurrences (across both survey periods)
- **≥ 7** occurrences in at least one survey period

Percent cover estimates and species identification

Percent cover estimates:



- % cover was estimated for each quadrat and averaged across all 4 quadrats

Species ID:

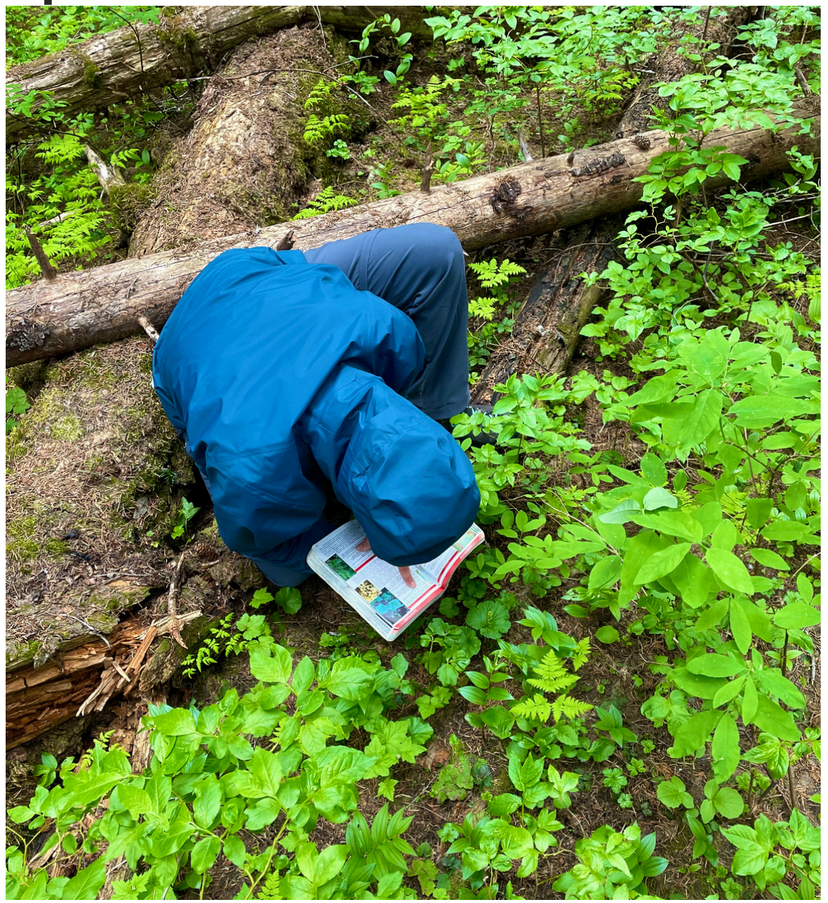
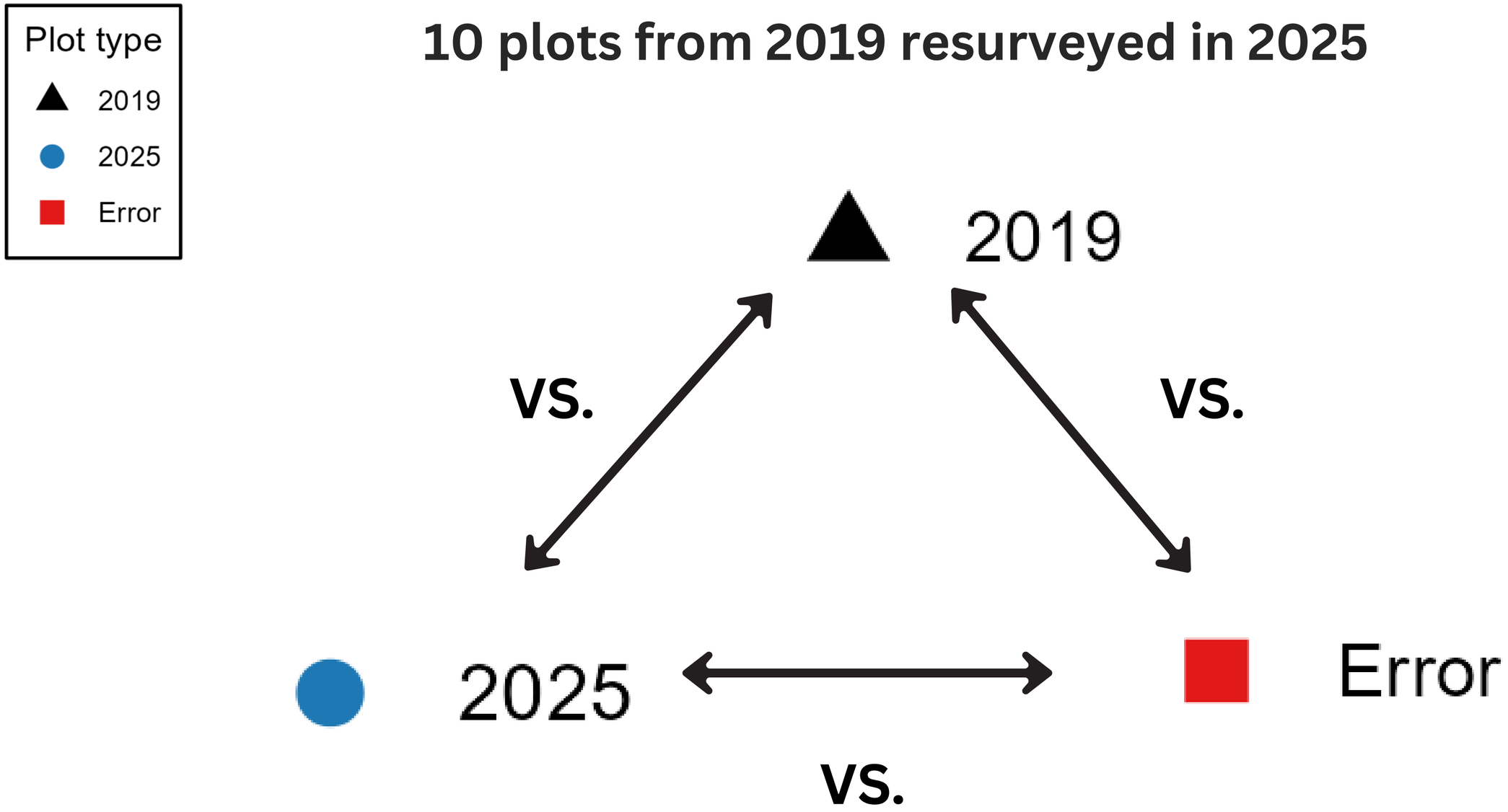


Photo: Rachel Wilson using the Pojar & MacKinnon (1994) to identify a *Mitella* species

Plant ID resources: Pojar & MacKinnon (1994), Parish et al. (1999), iNaturalist

10 plots from 2019 resurveyed in 2025



Summary and broader implications

Why it matters:

- Mountain ecosystems are warming rapidly, driving potential range shifts and community change

- Long-term resurveys allow us to detect ecological responses to climate change to better inform conservation policies and build better climate models